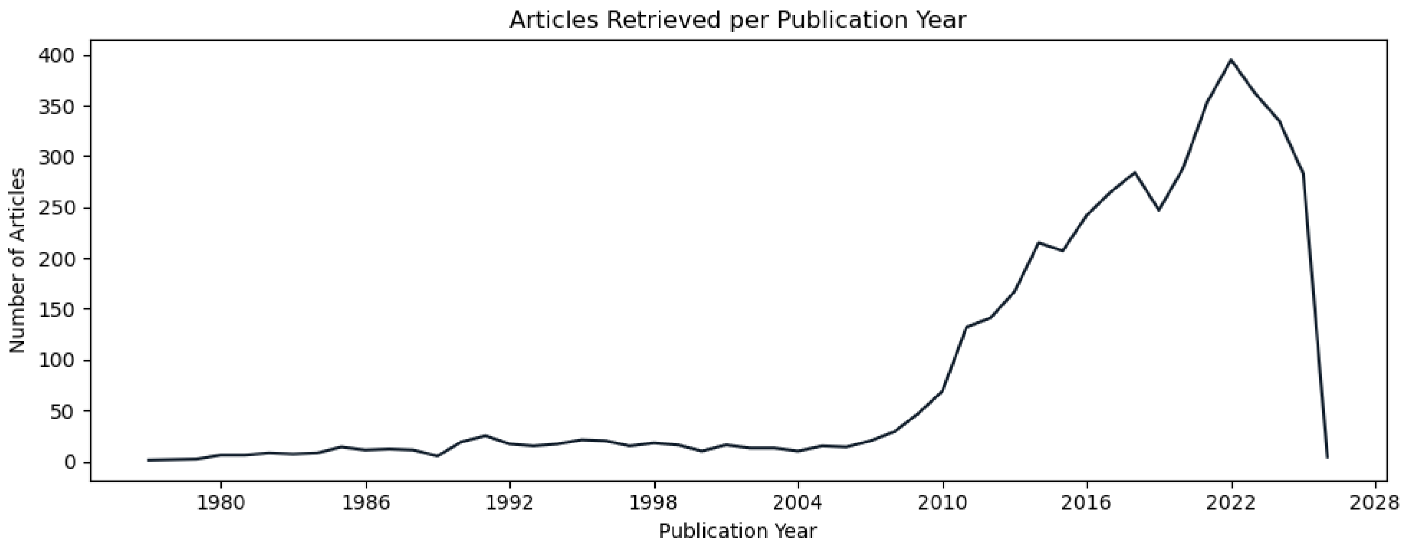
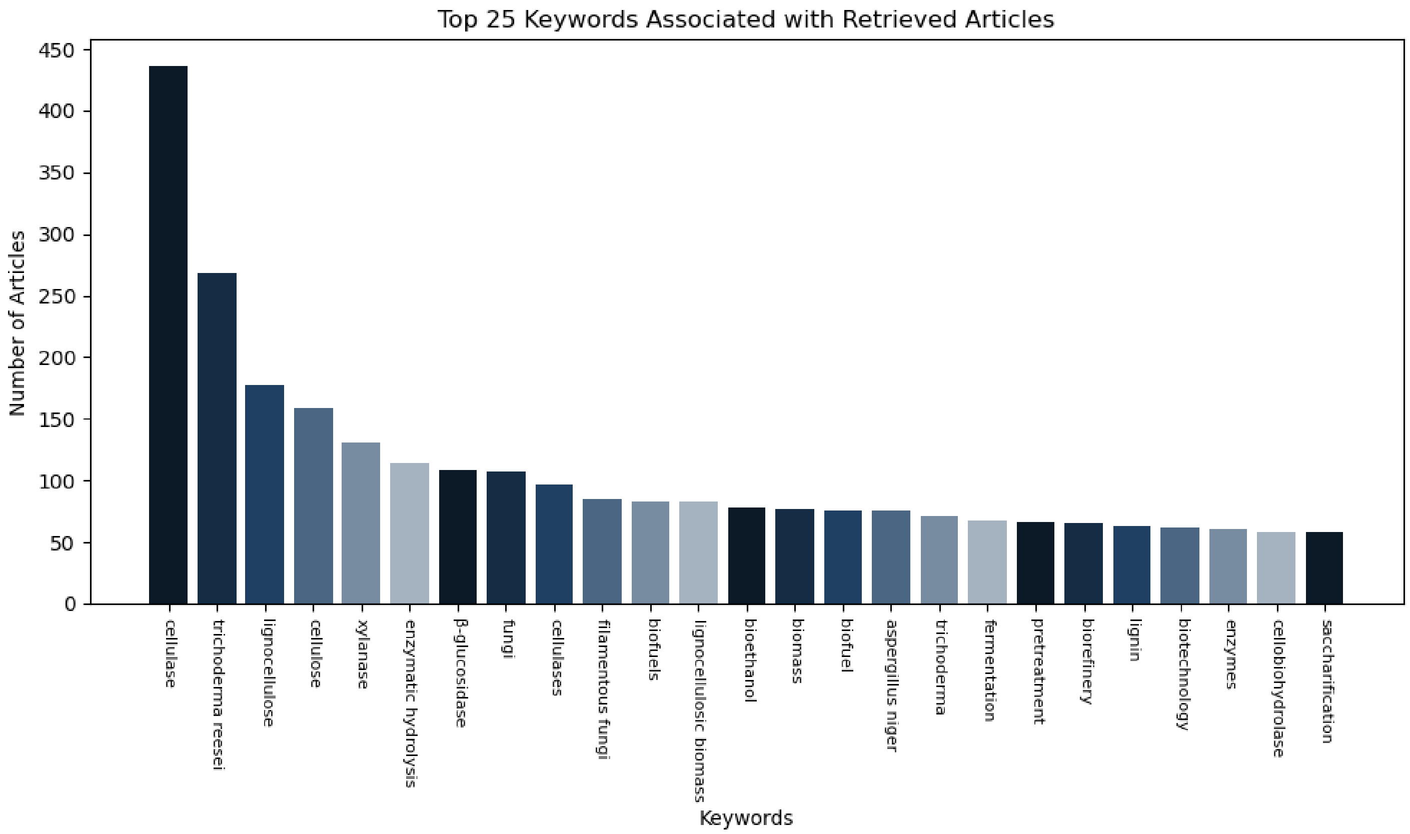
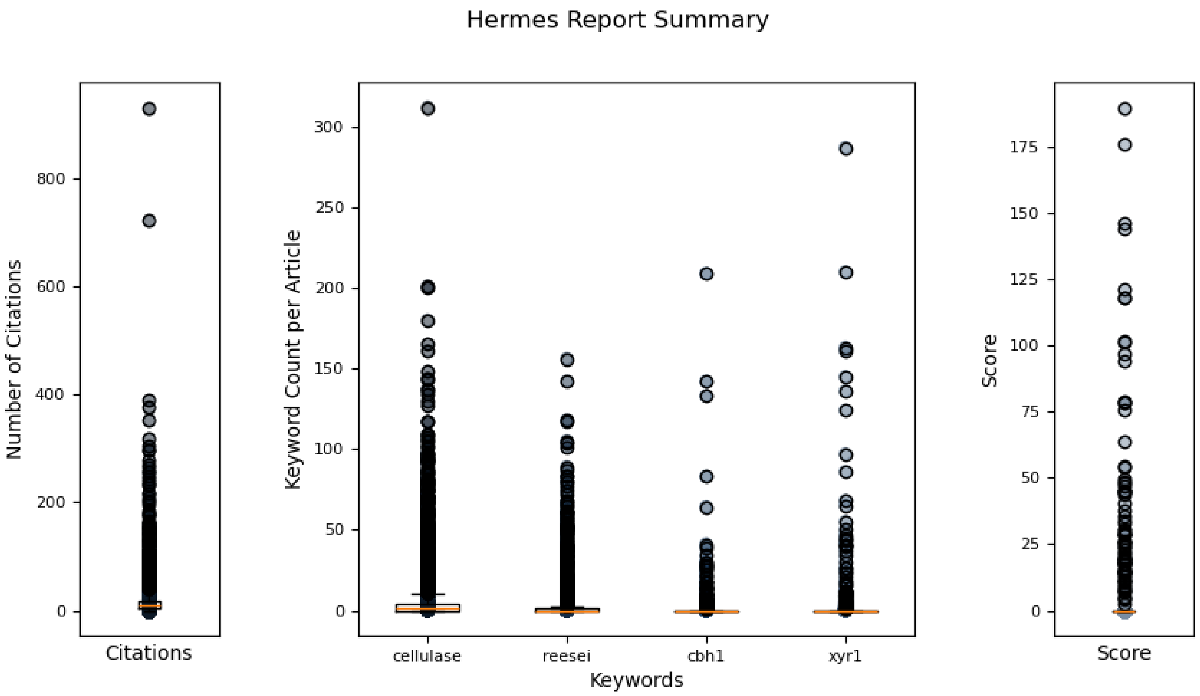
Literature Mining Report

Query Statistics

Query Date: 2025-12-19
Query Terms: "cellulase production trichoderma reesei"
Scoring Keywords: cbh1; cellulase; reesei; xyr1 (Strict Mode)
Articles Parsed: 4461/4461 (100.00%)
Keyword Rarity Score (IDF): xyr1: 3.285; cellulase: 0.665; reesei: 1.015; cbh1: 3.000







Query Results

PMID	Year	Journal	Authors	Title	Citations	Score
25909478	2015	BMC Genomics	Lichius, Alexan et al.	genome sequencing of the trichoderma reesei q...	23	189.84
25302561	2014	Molecular Micro	Lichius, Alexan et al.	nucleo-cytoplasmic shuttling dynamics of the ...	40	175.90
24294104	2013	Current Genomic	Amore, Antonell et al.	regulation of cellulase and hemicellulase gen...	86	146.22
35317826	2022	Microbial Cell	Mattam, Anu Jos et al.	factors regulating cellulolytic gene expressi...	22	144.01
32765463	2020	Frontiers in Mi	Jiang, Xianzhan et al.	improved production of majority cellulases in...	7	121.01
29079620	2017	Applied and Env	Kiesenhofer, Da et al.	influence of cis element arrangement on promo...	19	118.06
32877410	2020	PLoS Genetics	Zheng, Fanglin et al.	trichoderma reesei xyr1 activates cellulase g...	18	118.05
31517564	2020	RNA Biology	Till, Petra et al.	regulation of gene expression by the action o...	6	101.72
22080343	2011	Applied Microbi	Uzbas, Fatma et al.	a homologous production system for trichoderm...	43	101.39
36376415	2022	Scientific Repo	Arai, Toshiharu et al.	inducer-free cellulase production system base...	6	97.00
36528622	2022	Biotechnology f	Shen, Linjing et al.	tailoring the expression of xyr1 leads to eff...	3	94.37
34983541	2022	Microbial Cell	Wang, Yifan et al.	development of a powerful synthetic hybrid pr...	5	79.08
22750088	2013	Journal of Biot	Kubicek, Christ	systems biological approaches towards underst...	50	78.54
33606681	2021	PLoS Genetics	Wang, Lei et al.	interdependent recruitment of cyc8/tup1 and t...	11	78.24
23991059	2013	PLoS ONE	Wang, Mingyu et al.	a mitogen-activated protein kinase tmk3 parti...	44	75.77
38187093	2023	Synthetic and S	Zhao, Qinqin et al.	structure-guided engineering of transcription...	4	63.65
25626518	2015	Molecular Micro	Ghassemi, Sara et al.	the -importin kap8 (pse1/kap121) is required...	14	54.51
32671045	2020	Frontiers in Bi	Meng, Qing-Shan et al.	engineering the effector domain of the artifi...	3	54.10
30320082	2018	Frontiers in Bi	Alazi, Ebru et al.	modulating transcriptional regulation of plan...	15	49.83
35852347	2022	Microbiology Sp	Wang, Zhixing et al.	functional characterization of sugar transpor...	6	48.50
38548869	2024	Communications	Zhu, Zhihua et al.	the protein methyltransferase trsam inhibits ...	3	47.98
31077201	2019	Microbial Cell	Liu, Pei et al.	enhancement of cellulase production in tricho...	16	46.87
26360497	2015	PLoS Genetics	Li, Zhonghai et al.	synergistic and dose-controlled regulation of...	82	45.25
30902998	2019	Scientific Repo	Rantasalo, Anss et al.	novel genetic tools that enable highly pure p...	27	44.74
32695894	2020	Synthetic and S	Han, Lijuan et al.	redesigning transcription factor cre1 for all...	15	44.60

Query Result 1/25

Genome sequencing of the Trichoderma reesei QM9136 mutant identifies a truncation of the transcriptional regulator XYR1 as the cause for its cellulase-negative phenotype

Lichius, Alexander; Bidard, Frdrique; Buchholz, Franziska; Le Crom, Stphane; Martin, Joel; Schackwitz, Wendy; Austerlitz, Tina; Grigoriev, Igor V; Baker, Scott E; Margeot, Antoine; Seiboth, Bernhard; Kubicek, Christian P

BMC Genomics

Year: 2015 PMID: 25909478 Citations: 23 Score: 189.84 DOI: 10.1186/s12864-015-1526-0

Keywords Hits: cellulase: 49, reesei: 42, cbh1: 2, xyr1: 210

Keywords Frequency (TF): cellulase: 0.007; reesei: 0.006; cbh1: 0.000; xyr1: 0.031

Abstract:

Trichoderma reesei is the main industrial source of cellulases and hemicellulases required for the hydrolysis of biomass to simple sugars, which can then be used in the production of biofuels and biorefineries. The highly productive strains in use today were generated by classical mutagenesis. As byproducts of this procedure, mutants were generated that turned out to be unable to produce cellulases. In order to identify the mutations responsible for this inability, we sequenced the genome of one of these strains, QM9136, and compared it to that of its progenitor T. reesei QM6a.

Summary:

Background: Public concerns about the consequences of the continued production of fuels and chemicals from fossil carbon sources have today led to intensive attempts to switch to the use of renewable carbon resources for future energy supply, such as lignocellulosic biomass from agricultural crop residues, grasses, wood and municipal solid waste. This affects not only the production of ethanol as a biofuel but also a range of key biochemical building blocks for biorefineries, including 1,4-dicarboxy acids, 3-hydroxypropionic acid, levulinic acid, and polyols; to only name a few. Thereby, the costs of cellulases and hemicellulasses needed for hydrolysis of the plant biomass to soluble substrates contribute substantially to the price, and much cheaper sources of these enzymes are urgently required.

Results: The strain QM9136 has been isolated by a single round of X-ray mutagenesis using a linear particle accelerator from the wild type strain Q6a and was identified to be unable to produce cellulases . To get the complete picture of genetic modifications in that strain, we conducted whole genome comparisons between the natural isolate, the QM6a strain, and the mutant strain Qm9136 which we sequenced using high-throughput Illumina sequencing. To ...

Conclusion: The missing 140 C-terminal amino acids of XYR1 are therefore responsible for its previously observed auto-regulation which is essential for cellulases to be expressed. Our data present a working example of the use of genome sequencing leading to a functional explanation of the QM9136 cellulosae-negative phenotype. Electronic supplementary materialThe online version of this article (doi:10.1186/s12864-015-1526-0) contains supplementary materials, which are available to authorized users.

Discussion: Historically, the cloning of genes was based on the use of complementation of mutants, which favored those organisms which either contained plasmids or autonomously replicating elements (for easy recovering of the target gene) or which could efficiently be crossed in the laboratory. Although a number of substitute strategies had been developed (cf.), this nevertheless strongly limited the progress in organisms that exhibited neither of these possibilities, such as the so-called imperfect fungi. The advances in sequencing technologies and bioinformatics have now made it possible to sequence even full eukaryotic genomes quickly, and to use this strategy to identify mutations responsible for a particular phenotype -. However, since mutant organisms often contain several mutations (cc.) , most researchers either mapped the target mutation or backcrossed their mutant against the wild type, and sequenced more than one mutant genome simultaneously. Since the T. reeseimutants were maintained asexually, neither of those two strategies was possible for the present study.

Conclusions: Our study demonstrates how high-throughput genomic sequencing combined with reverse genetics can provide novel functional insights into organisms that are otherwise only poorly susceptible to mutational analyses. The mutation in xyr1inT. reeseiQM9136 also allowed new insights into the function of this central cellulase regulator. The ongoing investigation of yet two more cellullase-negative strains of T. reaesi (i.e. QM9978 and QM 9979;) which bear a functional xir1copy is expected to further elucidate celluluse and hemicellulases regulation in this important filamentous fungus.

Figures:

Figure 1: Pedigree of the cellulase-negative strain QM9136. UV light and chemical mutagenesis programs of the NATICK 3 , 4 , 9 and RUTGERS 27 series lead to the selection of T. reesei strains with improved (+) or impaired () cellulase production. Adapted with permission from 2 .

Figure 2: XYR1 structure and structural consequences of the truncated XYR 1780 . (A) Predicted coiled-coil domains in XYR1 using COILS, which compares a sequence to a database of known parallel two-stranded coiled-coils and derives a similarity score from which the probability that the sequence will adopt a coiled-coil confirmation is calculated. (B) XYR1 contains a classical tri-partite nuclear localisation signal (NLS; green) which directs the transcription factor into the nucleus. XYR1s DNA-binding domain (DBD; yellow) is orthologous to that of Gal4 from Saccharomyces

cerevisiae . The first coiled-coil domain of XYR1 (orange) is located at the end of the DBD. The filamentous fungal-specific transcription factor domain (FSTFD; blue) regulates target gene expression and constitutes the biggest part of the protein. A second coiled-coil domain (orange) overlaps with the C-terminal end of the FSTFD. The acidic activation domain (AAD; dark red) is a characteristic feature of Zn 2 Cys 6 -type transcription factors. The third coiled-coil domain (orange) sits at the very C-terminus, and likely mediates homodimerisation of XYR1. Hence, the truncated XYR1 1780 protein encoded in QM9136 almost completely lacks the AAD and fully lacks the third coiled-coil domain due to a frame-shift mutation from L765 onwards. The terminal 16 amino acids (L765-F780; light red) encode a non-sense sequence that finds no BLASTp hit in the T. reesei genome. (C) 3D-reconstruction of XYR1. RaptorX protein structure prediction software (raptorx.uchicago.edu) identifies three functional domains in XYR1, resembling DNA-binding domain (DBD), fungal-specific transcription factor domain (FSTFD) and acidic-activation domain (AAD), perfectly matching our manual annotation (B) . Five of the seven -helices comprising the AAD are missing in the truncated XYR1 1780 protein (highlighted in blue). The closest structural orthologs are the Zn 2 Cys 6 -type transcriptional regulators Ppr1 and Gal4 from S. cerevisiae , both known to be active as non-symmetrical homodimers 63 .

Figure 3: Gene replacement with full-length GFP-XYR1 construct rescues XYR1-loss-of-function phenotype of QM9136. (A) QM9136::GFP-XYR1 transformants (TRAL008 clones 1 to 3) grow on 65mM D-xylose show colony extension speeds comparable to the moderate producer strain QM9414 and QM9414::GFP-XYR1 transformants (TRAL002 clones 1 and 2). QM9136 and xyr1 are sensitive to 65mM D-xylose and display significantly reduced colony extension speeds. (B) Macroscopic colony phenotypes of GFP-XYR1 transformants TRAL002 and TRAL008, and control strains after 5days on MA medium supplemented with 65mM D-xylose. The XYR1-loss-of-function strains xyr1 and QM9136 develop a slower and less dense growing mycelium, however, start conidiation earlier, compared to the reference strain QM9414. Expression of GFP-XYR1 restores wild type morphology in QM9136 but does not alter QM9414. (C) Gene expression of the major cellulase cel7a , xylanase xyn2 and transcription factor xyr1 itself, induced by one hour exposure to 1.4mM sophorose, are restored in QM9136::GFP-XYR1 transformants (TRAL008). The XYR1-loss-of-function strains QM9136 and xyr1 show now transcriptional response to induction by sophorose. Both cellulase-positive controls, QM9414 and QM9414::GFP-XYR1 transformants (TRAL002) respond as expected with upregulated cel7a , xyn2 and xyr1 expression. Error bars show standard deviations (n=6 from three experimental and two biological replicates).

Figure 4: Gene replacement with the truncated GFP-XYR1 1780 reproduced the XYR1-loss-of-function phenotype in QM9414. (A) QM9414::GFP-XYR1 1780 transformants (TRAL007 clones 1 to 3) showed the same growth defect on 65mM D-xylose as QM9136 and xyr1 , respectively, indicating loss of XYR1 function, which was further verified by the corresponding macroscopic colony phenotypes, shown in (B) . (C) Gene expression of cellulase cel7a , xylanase xyn2 and xyr1 were not significantly upregulated in GFP-XYR1 1780 expressing transformants (TRAL007), whereas the GFP-XYR1 transformants (TRAL002) clearly responded to the inductive stimulus provided by sophorose. Error bars show standard deviation (n=6 from three experimental and two biological replicates involving two different TRAL007 clones).

Figure 5: Nuclear import of GFP-XYR1 is indistinguishable in QM9136 and QM9414 transformants. (A) Sophorose-induced nuclear import of GFP-XYR1 in QM9136 and QM9414 backgrounds was indistinguishable. Relative fluorescence intensity represents the average GFP-XYR1 signal detected measured in 120 nuclei per sample. The nucleo-cytoplasmic ratio (n/c-ratio) provides a measure of how much brighter the average GFP-XYR1 signal is inside nuclei compared to the surrounding cytoplasm. Error bars represent standard deviation (n=120). (B) Representative, inverted fluorescence images showing increasing GFP-XYR1 recruitment into nuclei of T. reesei germlings over a time course of 60min after carbon source replacement with 1.4mM sophorose.

Figure 6: Western blot analysis of GFP-XYR1 and GFP-XYR1 1780 under cellulase inducing conditions. Top row: Coomassie stained loading control; middle row: chemifluorescent image of Western blot membrane; bottom row: x-ray image of Western blot membrane. (A) With time (1-3h of sophorose induction), less of the 127kDa full-length GFP-XYR1(arrowheads) but more of its various 30-70kDa degradation products (asterisks) can be detected in protein extracts. This rapid turn-over of the transcription factor is conserved in QM9414 and QM9136 transformants. (B) In comparison to full-length GFP-XYR1 (arrowheads) expressed in QM9414 and QM9136 transformants, of the truncated GFP-XYR1 1780 construct, only 80kDa or smaller degradation products could be detected (asterisk). 5-30g refer to the total protein load per lane; numbers on the side denote molecular weight ranges; M1 is PageRuler pre-stained molecular weight marker; M2 is SuperSignal enhanced protein ladder; 18h QM6a GFP is the positive control, and 18h QM9414 the negative control for GFP(B-2) antibody specificity.

Figure 7: C-terminal truncation renders GFP-XYR1 1780 non-responsive to inductive signals. (A) Nuclear recruitment dynamics assessed by quantitative live-cell imaging. In contrast to the full-length GFP-XYR1 (TRAL002), whose concentration in nuclei rapidly increased by a factor of up to eight-fold, the truncated GFP-XYR1 1780 (TRAL007) was unable to significantly increase its presence in the nucleus in response to cellulase- and xylanase-inducing carbon sources (1.4mM sophorose, 25mM lactose and 1mM D-xylose). Instead, GFP-XYR1 1780 consistently showed nuclear presence comparable to the basal presence of GFP-XYR1 under xyr1 -non-inducing conditions (65mM D-xylose). Error bars show standard deviation (n=2 from two biological replicates involving two different TRAL007 clones). (B) Representative, inverted fluorescence images showing nuclear import of GFP-XYR1 and GFP-XYR1 1780 after one hour of sophorose induction. In contrast to the full-length GFP-XYR1 construct, the truncated version is barely visible inside nuclei before induction (arrows). Most importantly, upon addition of 1.4mM sophorose, the nuclear signal of GFP-XYR1 1780 does

increase, however, only to a fraction of the intensity (about 15%) of the GFP-XYR1 construct under identical conditions. Scale bar, 10m.

Figure 8: Quantification of nuclear recruitment of GFP-XYR1 and GFP-XYR1 1780 in the three main functional zones of the mature colony. All strains were grown for 48hours on MA medium agar plates supplemented with the indicated carbon source. (A) Full-length GFP-XYR1 responds to different xyr1 -inducing signals with nuclear recruitment, most prominently in the central area of the colony, and dependent on the inductive strength of the provided carbon source (25mM lactose1mM D-xylose1% (w/v) cellulose). Under non-inducing/repressing conditions (50mM D-glucose or 65mM D-xylose) only weak baseline nuclear recruitment can be detected. In contrast, the truncated GFP-XYR1 1780 does not significantly respond to cellulase induction, and hence can only be detected at baseline levels independent of the provided carbon source. (B) Representative, inverted fluorescence images of the tested conditions, showing that GFP-XYR1 1780 does respond to 25mM lactose with weak but detectable nuclear import, comparable to the baseline presence of GFP-XYR1 under repressing conditions (barely visible nuclei are indicated with arrows). On 50mM D-glucose GFP-XYR1 1780 is visible in small spots, suggesting degradation of the non-functional XYR1 construct in the proteasome.

Mentioned biomedical entities:

Genes: ACE2, ACE3, AM, B, CBM1, CDC1, CRE1, ENS, Gal4, MA, Ppr1, RUT, SEB, SRA, XYR1, XlnR, Xyr1, cbh1, cel6a, cellulase, cre1, ptrA, tef1, xyn1, xyn2, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, cell

Organisms: GE

Tissues: lens, tube

Pathways: N/A

Query Result 2/25

Nucleo-cytoplasmic shuttling dynamics of the transcriptional regulators XYR1 and CRE1 under conditions of cellulase and xylanase gene expression in Trichoderma reesei

Lichius, Alexander; Seidl-Seiboth, Verena; Seiboth, Bernhard; Kubicek, Christian P

Molecular Microbiology

Year: 2014 PMID: 25302561 Citations: 40 Score: 175.90 DOI: 10.1111/mmi.12824

Keywords Hits: cellulase: 74, reesei: 29, cbh1: 9, xyr1: 163

Keywords Frequency (TF): cellulase: 0.009; reesei: 0.003; cbh1: 0.001; xyr1: 0.019

Abstract:
Trichoderma reesei is a model for investigating the regulation of (hemi-)cellulase gene expression. Cellulases are formed adaptively, and the transcriptional activator XYR1 and the carbon catabolite repressor CRE1 are main regulators of their expression. We quantified the nucleo-cytoplasmic shuttling dynamics of GFP-fusion proteins of both transcription factors under cellulase and xylanase inducing conditions, and correlated their nuclear presence/absence with transcriptional changes. We also compared their subcellular localization in conidial germlings and mature hyphae. We show that cellulase gene expression requires de novo biosynthesis of XYR1 and its simultaneous nuclear import, whereas carbon catabolite repression is regulated through preformed CRE1 imported from the cytoplasmic pool. Termination of induction immediately stopped cellulase gene transcription and was accompanied by rapid nuclear degradation of XYR1. In contrast, nuclear CRE1 rapidly decreased upon glucose depletion, and became recycled into the cytoplasm. In mature hyphae, nuclei containing activated XYR1 were concentrated in the colony center, indicating that this is the main region of XYR1 synthesis and cellulase transcription. CRE1 was found to be evenly distributed throughout the entire mycelium. Taken together, our data revealed novel aspects of the dynamic shuttling and spatial bias of the major regulator of (hemi-)cellulase gene expression, XYR1, in T. reesei.

Summary:
Introduction: In nature fungi contribute as essential decomposers of complex organic molecules to the biological carbon cycle. These organic polymers, produced through carbon dioxide fixation by plants, mainly comprise plant cell wall polysaccharides including cellulose, hemicelluloses, pectins and the polymer lignin. The major plant cellwall component is the -(1,4)-linked glucose polymer, cellulose that alone exhibits an annual production of 7.2 10 10 tons, and its microbial degradation is therefore a key transformation step in the biological Carbon Cycle.

Results: XYR1 requires N-terminal and CRE1 requires C-terminum GFP labeling in order to preserve protein function. Consequently, only T. reeseitransformants expressing GFPXYr1 and Cre1GFP, respectively, were used for all subsequent experiments. Xyr1 requires GFP and CRe1 requires c-terminus GFP. xyr1/cre1

Discussion: It is commonly believed that most Zn2Cys6zinc cluster proteins are localized within the nucleus on a constitutive and permanent basis (MacPherson However, there is now an increasing number of these zinc cluster proteins from multicellular fungi known, which do not share this behavior (Shimizu Berger Makita Dinamarco Here we showed that the XYR1 transcription factor ofT. reeseialso joins this list.

Figures:
Figure 1: Nuclear import of XYR1 was significantly increased under cellulase inducing conditions.A and B. (A) GFPXYR1 became strongly recruited into nuclei in the presence of lactose, whereas it was only weakly expressed under repressing/non-inducing conditions on glucose (B).C and D. CRE1 did not show this specific response and appeared equally recruited to nuclei under cellulase inducing and repressing conditions. 3D surface plots (right row) of the inverted fluorescence images (middle row) illustrate the relative differences of localized fluorescence signal intensities. The corresponding transmission light images are shown in the left row. The displayed strains were incubated for 48h at 28C on MA agar plates supplemented with lactose or glucose respectively.Scale bars, 5m.

Figure 2: Protein turn-over kinetics of both fusion reporters differ, leading to significant nuclear accumulation of free GFP from GFPXYR1 degradation.A. Coomassie-stained SDS-PAGE gel. Except for the GFP positive control, of which only 0.35g total protein extract of QM6a P tef1 :: gfp were loaded, 20g of all other protein extract samples were loaded per lane. Molecular weight marker on both sides is PageRuler 10170kDa ladder.B. Western blot of the sister SDS-PAGE gel shown in (A), incubated with monoclonal -GFP(B-2) antibody, ECL 2 reagent and imaged on a Typhoon FLA700 chemifluorescence imager. Molecular weight marker on the left (lane 1) is PageRuler 10170kDa ladder, on the right (lane 8) is SuperSignal Enhanced 20 150kDa ladder. Cytosolic GFP (26.98 kDa) as positive control is exclusively detected in lane 2. Full-length CRE1GFP (71.07kDa) bands in lane 3 just above the 70kDa marker, and some of its degradation products with decreasing abundance towards low-molecular-weight species below. Lanes 46 show full-length GFPXYR1 (128.41kDa) with decreasing abundance after 13h of sophorose induction. Interestingly, with time, abundance of its higher-molecular-weight degradation products (4080kDa) decreases, while amounts of its low-molecular-weight degradation products (25 30kDa) significantly increase. The negative control in lane 7 (protein extract from an 18h glucose culture of QM9414 t ku70) reveals minor non-specific background detection in the higher molecular weight range40kDa.C. The same membrane as shown in (B) developed by x-ray film exposure, shows identical results with less non-specific background signals, however, inferior spatial resolution. Intense nuclear fluorescence of GFPXYR1 and

CRE1GFP, respectively, was confirmed by live-cell imaging in all samples prior to biomass harvest and protein extraction.

Figure 3: XYR1 and CRE1 showed distinct nuclear shuttling dynamics. Inverted fluorescence images show the degree of nuclear transcription factor recruitment at representative time points, with the corresponding quantitative analysis below.A. Time-course of sophorose-induced nuclear import of GFPXYR1. Quantitative image analysis showing immediate but weak nuclear recruitment which started to significantly increases after 30min upon induction, probably correlated with the maturation time of newly synthetized GFP.B. Time-course of glucose-induced nuclear loss of GFPXYR1. Quantitative image analysis showing only slow decrease of nuclear XYR1 signal within the first hour upon carbon source replacement.C. Time-course of glucose-induced nuclear import of CRE1GFP. Quantitative image analysis showing rapid nuclear import of CRE1 from the cytoplasmic pool.D. Time-course of sophorose-induced nuclear loss of CRE1GFP. Quantitative image analysis showing rapid nuclear export of CRE1 upon cellulase induction with sophorose. Please refer to the text for more detailed explanation.Scale bars, 5m.

Figure 4: XYR1 nuclear import dynamics correlated with gene expression.A. Sophorose-induced nuclear import of XYR1 represents cellulase inducing conditions.B. Low-xylose (1mM) induced nuclear import of XYR1 represents cellulase and xylanase inducing conditions.C. High-xylose (65mM) induced XYR1 suppression represents carbon-catabolite repressing conditions.D. Starvation-induced nuclear import of XYR1 represents carbon-catabolite de-repressing conditions.E. Gene expression analysis from biomass samples collected at the end of each imaging experiment quantified in (AD) showed that upon sophorose addition cellulase cbh1 and xyr1 itself were strongly upregulated, low concentration (1mM) of xylose strongly induce cbh1 and xyn2 , but not xyr1 , whereas in the presence of 65mM xylose none of these three marker genes was significantly upregulated. This is different under de-repressing conditions, i.e. in the absence of any experimentally added carbon source, where nuclear import of XYR1 evidently induced moderate upregulation of cbh1 , xyn2 and of itself.

Figure 5: Complete inhibition of XYR1 nuclear import and cellulase gene expression by cycloheximide treatment. (A)(D) show the shuttling dynamics of fluorescently labeled XYR1 and CRE1 upon cellulase induction and repression, in the presence and absence of 50M cycloheximide (CHX) respectively. (E) and (F) show the corresponding transcriptional analyses.A. Addition of CHX completely blocked de novo protein biosynthesis of XYR1 and consequently its nuclear import.B. Nuclear loss of CRE1 remained unaffected under the same conditions.C. Nuclear loss of XYR1, on the other hand, was slightly enhanced in the presence of CHX in comparison to the untreated control.D. Glucose-induced nuclear import of CRE1 showed no significant differences between CHX-treated sample and untreated control.E. Inhibition of de novo XYR1 protein biosynthesis and nuclear import was directly correlated to the inhibition of transcription of XYR1-dependent genes (cbh1 and xyn2). Transcription of xyr1 itself, however, responded normally to the induction signal and became upregulated to the same extend as in the untreated controls.F. Apart from a weak upregulation of cre1 in the presence of CHX, all other tested marker genes became downregulated as expected under cellulase non-inducing conditions.

Figure 6: Transcription factor recruitment in the functionally stratified colony.A. Forty-eight-hour lactose plate culture of a GFPXYR1 expressing strain. The white dotted rectangle indicates the sampling area which includes all three main functional zones of the colony (central area, subperiphery and leading edge) assessed by live-cell imaging microscopy.B. Bright-field images show the typical hyphal phenotypes: dense hyphal network with conidiation (conidia indicated with asterisks) and a high degree of vacuolation (vacuoles indicated with arrowheads) in the central area, branched hyphae with beginning vacuolation but no conidiation in the subperiphery, and largely non-vacuolated leading hyphae with clear apical nuclear-exclusion zone (indicated with black lines). Representative images of the GFPXYR1 fluorescence signal from these three areas show the evident decrease in nuclear XYR1 signal from colony center to colony edge. Scale bars, 5m.C and D. Quantification of nuclear recruitment of XYR1 and CRE1, respectively, in these three zones on inducing (cellulose and lactose) and non-inducing/repressing (glucose and xylose) carbon sources showed that nuclear import of XYR1 peaked in the central colony area exclusively under cellulase inducing conditions. CRE1, on the other hand, was equally recruited to nuclei in all regions of the colony with no significant differences between the tested carbon sources. All cultures have been incubated for 48h at 28C on MA medium supplemented with 1% w/v of the respective carbon source.

Mentioned biomedical entities:

Genes: ACE1, ACE2, AflR, CRE1, Cellulase, Cre1, CreA, Gal1, Gal4, Leu3, Lys14, MA, Mig1, Oaf1, Put3, Snf1, Tel, War1, XYR1, XlnR, Xyr1, cbh1, cellulase, cre1, gpd1, hydrolase, pkiA, tef1, xyn1, xyn2, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, cell

Organisms: A. nidulans, E. coli, GE

Tissues: lens, pulp, tube

Pathways: N/A

Query Result 3/25

Regulation of Cellulase and Hemicellulase Gene Expression in Fungi

Amore, Antonella; Giacobbe, Simona; Faraco, Vincenza

Current Genomics

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Keywords Hits: cellulase: 180, reesei: 50, cbh1: 15, xyr1: 30

Keywords Frequency (TF): cellulase: 0.019; reesei: 0.005; cbh1: 0.002; xyr1: 0.003

Abstract:
Research on regulation of cellulases and hemicellulases gene expression may be very useful for increasing the production of these enzymes in their native producers. Mechanisms of gene regulation of cellulase and hemicellulase expression in filamentous fungi have been studied, mainly in Aspergillus and Trichoderma. The production of these extracellular enzymes is an energy-consuming process, so the enzymes are produced only under conditions in which the fungus needs to use plant polymers as an energy and carbon source. Moreover, production of many of these enzymes is coordinately regulated, and induced in the presence of the substrate polymers. In addition to induction by mono- and oligo-saccharides, genes encoding hydrolytic enzymes involved in plant cell wall deconstruction in filamentous fungi can be repressed during growth in the presence of easily metabolizable carbon sources, such as glucose. Carbon catabolite repression is an important mechanism to repress the production of plant cell wall degrading enzymes during growth on preferred carbon sources. This manuscript reviews the recent advancements in elucidation of molecular mechanisms responsible for regulation of expression of cellulase and hemicellulase genes in fungi.

Summary:
INTRODUCTION: Cellulase enzymes are required for the hydrolysis of polysaccharides in lignocellulosic biomass. The enzymes involved in the hydrolase of polymers are: -glucosidase, -1,4-xylanases, xylosidasses, endo--1-1, 4-xylases and -arabinofuranosidas. The enzymatic activities involved in cellulose hydrolysis are:

CONCLUSIONS: The regulation of (hemi)cellulolytic genes appears to be basically the same among filamentous fungi such as T. reesei, N. crassa, Asperigillusspp., although their regulatory mechanisms are quite complex and present some differences. In particular, cross talks between expression of cellulolysis and hemicellulase genes make the regulatory mechanisms more complicated. Xyr homologs (Xy1 ofT. reaesi, XlR ofN. aspergillus, and XnR ofAspergillusspp) mediate expression of both xylanolytic and cellulalytic genes in response to xilan.

Figures:
Figure 1: Schematic representation of transcriptional factors affecting cellulases and xylanases expression in T. reseei (black box), N. crassa (grey box) and Aspergillus spp. (white box). The carbon catabolite repressor CRE, the activators clbR, Xyr/xlR/XlnR,Clr,ACE2, the repressorACE1, the CCAAT binding Hap2/3/5 complex, the pH regulator PacC, and the nitrogen regulators AreA and Nit2 are shown. The repressionactivity ( ), the induction activity ( ) and also promoter ( ) and coding region ( ) are indicated.

Mentioned biomedical entities:
Genes: ACE1, ACE2, Ace1, Ace2, AceI, AreA, CBH, CEL3A, CEL5A, CEL6A, CEL7A, CLR-1, CPC1, CRE, CRE-1, CRE1, Cellulase, ClbR, CreA, CreB, CreD, GATA-1, HAP2, HAP3, HAP5, Hap, LAE1, LIM1, NF-YA, NFYB, NIT2, Nit2, PY, PacC, RUT, SNF1, XYR1, XlnR, Xyr1, abfB, ace1, ace2, areA, cbh1, cbh2, cel6a, cellulase, clbR, cmc1, cre-1, cre1, cre2, creA, egl1, gh10-2, gsn, hap2, histone, nag1, pelA, pmeA, swo1, xlnD, xlnR, xyn1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: A. aculeatus, A. fumigatus, A. nidulans, A. phoenicis, A. terreus, human

Tissues: N/A

Pathways: N/A

Query Result 4/25

Factors regulating cellulolytic gene expression in filamentous fungi: an overview

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Microbial Cell Factories

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Keywords Hits: cellulase: 200, reesei: 105, cbh1: 17, xyr1: 68

Keywords Frequency (TF): cellulase: 0.023; reesei: 0.012; cbh1: 0.002; xyr1: 0.008

Abstract:

The growing demand for biofuels such as bioethanol has led to the need for identifying alternative feedstock instead of conventional substrates like molasses, etc. Lignocellulosic biomass is a relatively inexpensive feedstock that is available in abundance, however, its conversion to bioethanol involves a multistep process with different unit operations such as size reduction, pretreatment, saccharification, fermentation, distillation, etc. The saccharification or enzymatic hydrolysis of cellulose to glucose involves a complex family of enzymes called cellulases that are usually fungal in origin. Cellulose hydrolysis requires the synergistic action of several classes of enzymes, and achieving the optimum secretion of these simultaneously remains a challenge. The expression of fungal cellulases is controlled by an intricate network of transcription factors and sugar transporters. Several genetic engineering efforts have been undertaken to modulate the expression of cellulolytic genes, as well as their regulators. This review, therefore, focuses on the molecular mechanism of action of these transcription factors and their effect on the expression of cellulases and hemicellulases.

Summary:

Introduction: The gradual depletion of conventional fossil fuels and increasing awareness about the effects of greenhouse gas emissions has led to global interest in the development of renewable energy, particularly biofuels such as ethanol. Biofuels can be categorized into 1st, 2nd, 3rd, and 4th generation biofuel based on the feedstock used for production. The 1sth generation Biofuel are produced by the fermentation of sugar or starchy substrates and these processes have already been commercially established. However, considering the huge demand for ethanol, it has become necessary to expand the scope of fermentable substrates beyond the conventional ones (e.g. cane molasses, starch, etc.) to lignocellulosic biomass, which is available in abundance.

Conclusion: This review provides an overview of the complex regulatory network involved in the expression of cellulase and hemicellulases genes in filamentous fungi. It provides insights into the regulatory factors involved in this process and provides more targets for the rational engineering of filamentous filamentous fungal strains to develop better strains, for the production of cellulose and hemicellulose enzymes. The on-site production of cellsulased enzymes seems to be the most feasible solution to overcome the challenge of using expensive commercial cellulolytic enzymes for the hydrolysis of biomass, in 2G ethanol biorefineries. However, filamentous plants like T. reesei, P. funiculosum, Ps. oxalicum etc. that can produce cellullases and hemiscellullases have several inherent limitations in their enzyme production and secretion capacity. These can be overcome by genetic engineering to develop strains that produce enzymes at an industrially relevant scale.

Figures:

Figure 1: Transcriptional regulators and transporters involved in cellulase expression in T. reesei and Penicillium sp

Figure 2: Comparative analysis of genome with respect to CAZymes in Trichoderma reesei , Penicillium funiculosum and Penicillium oxalicum 1142. GHs Glycoside hydrolases, GTs Glycosyl transferase, PLs Polysaccharide Lyases, CEs carbohydrate esterases, AAS auxiliary activities, and CBMs carbohydrate binding modules

Mentioned biomedical entities:

Genes: 586, Ace1, Ace2, Ace3, AceI, Are1, AreA, Azf1, B, BglR, C, CYC8, Cdt1, CdtC, Clr1, Clr2, ClrB, Cre1, Cre2, CreA, CreB, CreC, Crt1, Crz1, Ctf1, GH, GHs, Gal11, HapB, LaeA, MFS, Pac1, PacC, PalA, STP1, Stp1, TUP1, TacA, Vel2, Vel3, VelB, Vib1, XlnR, Xyr1, ada, amdS, bgl1, bgl2, cbh1, cbh2, cellulase, creA, eg1, egl1, egl3, hydrolase, pac1, pdc, tubB, xlnR, xyn1, xyn2, xyn3, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, cell

Organisms: A. nidulans, Lantana, Pseudotsuga, T. reesei

Tissues: N/A

Pathways: N/A

Query Result 5/25

Improved Production of Majority Cellulases in Trichoderma reesei by Integration of cbh1 Gene From Chaetomium thermophilum

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Keywords Frequency (TF): cellulase: 0.009; reesei: 0.015; cbh1: 0.027; xyr1: 0.001

Abstract:
Lignocellulose is an abundant waste resource and has been considered as a promising material for production of biofuels or other valuable bio-products. Currently, one of the major bottlenecks in the economic utilization of lignocellulosic materials is the cost-efficiency of converting lignocellulose into soluble sugars for fermentation. One way to address this problem is to seek superior lignocellulose degradation enzymes or further improve current production yields of lignocellulases. In the present study, the lignocellulose degradation capacity of a thermophilic fungus Chaetomium thermophilum was firstly evaluated and compared to that of the biotechnological workhorse Trichoderma reesei. The data demonstrated that compared to T. reesei, C. thermophilum displayed substantially higher cellulose-utilizing efficiency with relatively lower production of cellulases, indicating that better cellulases might exist in C. thermophilum. Comparison of the protein secretome between C. thermophilum and T. reesei showed that the secreted protein categories were quite different in these two species. In addition, to prove that cellulases in C. thermophilum had better enzymatic properties, the major cellulase cellobiohydrolase I (CBH1) from C. thermophilum and T. reesei were firstly characterized, respectively. The data showed that the specific activity of C. thermophilum CBH1 was about 4.5-fold higher than T. reesei CBH1 in a wide range of temperatures and pH. To explore whether increasing CBH1 activity in T. reesei could contribute to improving the overall cellulose-utilizing efficiency of T. reesei, T. reesei cbh1 gene was replaced with C. thermophilum cbh1 gene by integration of C. thermophilum cbh1 gene into T. reesei cbh1 gene locus. The data surprisingly showed that this gene replacement not only increased the cellobiohydrolase activities by around 4.1-fold, but also resulted in stronger induction of other cellulases genes, which caused the filter paper activities, Azo-CMC activities and -glucosidase activities increased by about 2.2, 1.9, and 2.3-fold, respectively. The study here not only provided new resources of superior cellulases genes and new strategy to improve the cellulase production in T. reesei, but also contribute to opening the path for fundamental research on C. thermophilum.

Summary:

Introduction: Plant biomass from agriculture and forestry is one of the most abundant resources on the earth and it has been considered as a promising material for producing renewable biofuels and other value-added bio-products. One of the bottlenecks of this process is the enzymatic degradation of biomass-derived polysaccharides. Currently, the filamentous fungi Trichoderma, Aspergillus, and Penicillium species are the most commonly used producer for lignocellulases in industry. However, the optimum temperature of enzymes from these species is normally from 30 to 50C at which the efficiency of biomass polysacharides saccharification is very low.

Results: Chaetomium thermophilumexhibited greater utilization efficiency of cellulose materials compared to that ofT. reesei. The results suggest that the cellulose degradation capacity ofC. thermophiliumwas evaluated and compared with that of T. reaesi. To alleviate potential effects caused by differential spore germination efficiency,C. sporulation was firstly grown in medium containing glucose as the sole carbon source and the same amount of wet mycelia (about 2.2 g) were then transferred into medium based on Avicel or sugarcane bagasse as the carbon source. The abundance of biomass accumulation, the extracellular protein concentration, and the filter paper activities in the supernatant were measured at different time points.

Discussion: This study demonstrates that a thermophilic fungus C thermophilumsecreted fewer cellulases, but showed higher efficiency to degrade celluloses. This finding indicated that C. thermophiliummight contain cellulolytic enzymes with higher specific activities. However, UPLC-MS/MS analysis of C.thermophilumsupernatants demonstrated there were many oxidoreductases included in C. thermalophilumprotein secretome.

Conclusion: To search for new resources of lignocellulases for plant biomass degradation, here we reported that a thermophilic fungus C. thermophilum could degrade celluloses more efficiently with less secreted cellulase compared to T. reesei, implying the existence of excellent cellulolytic genes in its genome. Comparison of the enzyme properties of CBH1 from C. tropphilum and T.reespi further confirmed this hypothesis. More interesting, our data showed that only raising the function of C. reaesi could lead to a marked increase in the production levels of other cellullases. Based on the observed phenomenon, we speculated that if celluloids are accumulated temporarily, they might play an important role in the process of celluleases genes induction.

Figures:

Figure 1: Phenotypic analysis of C. thermophilum and T. reesei when response to cellulose. (A) The intracellular protein concentration of C. thermophilum and T. reesei mycelia collected from the culture under Avicel or sugarcane bagasse after transferring from the glucose culture. (B) Growth phenotype of C. thermophilum and T. reesei under Avicel or sugarcane bagasse. About 2.2 g wet mycelia of C. thermophilum and T. reesei

were collected and transferred into Avicel or sugarcane bagasse media from glucose contained media and cultivated for 48 h. (C) Total secreted protein levels in the supernatants of C. thermophilum and T. reesei in Avicel or sugarcane bagasse media at different time points. (D) Filter paper activities of the supernatants of C. thermophilum and T. reesei in Avicel or sugarcane bagasse media at different time points. Shown in (A , C,D) are the mean values of three biological replicates. Error bars show the standard deviations between these replicates. T. reesei -A, T. reesei-S , C. thermophilum -A, and C. thermophilum -S respectively indicates T. reesei grown on Avicel (A) or sugarcane bassage (S) and C. thermophilum grown on Avicel (A) or sugarcane bassage (S). Shown in (B) is one of the three represents.

Figure 2: The extracellular protein categories in the supernatants of cellulose cultures were different between C. thermophilum and T. reesei. (A) SDS-PAGE analysis of the secreted proteins in the supernatant of C. thermophilum and T. reesei in 120-h Avicel cultures after transferring from glucose cultures. (B) Major protein categories of the secretome of C. thermophilum and T. reesei under Avicel. The data was collected by UPLC-MS/MS analysis and the detailed information was listed in Supplementary Tables S2 S7 .

Figure 3: Expression of Tr-CBH1 and Ct-CBH1 in T. reesei using cellulase-free background system. (A) Schematic diagram of the construction of recombinant strain Tr-cT cbh1 . (B) Schematic diagram of the construction of recombinant strain Tr-cC cbh1 . FgDNA-P indicates fragments in the parental strain. FvDNA indicates Fragment from plasmid vector constructed for the generation of the indicated recombinant strain. FgDNA-T indicates DNA fragments in the generated transformant strain.

Figure 4: Ct-CBH1 showed higher specific activity compared to Tr-CBH1. (A) SDS-PAGE analysis of the secreted proteins in the supernatant of the recombinant T. reesei strains C pyr4 , Tr-cT cbh1 , and Tr-cC cbh1 . A suspension of 10 6 conidia/mL was inoculated into MM media with glucose as the only carbon source and cultivated for 48 h. A total of 20 L of the supernatant was used for SDS-PAGE analysis. (B) Total secreted protein levels and CBH activities in the supernatants of the indicated T. reesei strains. The samples used in (B) were as same as in (A) . Shown are the mean values of three biological replicates. Error bars show the standard deviations between these replicates. The significance of differences between Tr-cT cbh1 and Tr-cC cbh1 strains was based on t -test analysis by the FDR approach. Asterisks indicate significant differences (P 0.001). ns, not significant.

Figure 5: Ct-CBH1 showed higher optimum temperature and better temperature stability compared to Tr-CBH1. (A) Optimum pH determination of Tr-CBH1 and Ct-CBH1. (B) pH stability analysis of Tr-CBH1 and Ct-CBH1. (C) Optimum temperature determination of Tr-CBH1 and Ct-CBH1. (D) Temperature stability analysis of Tr-CBH1 and Ct-CBH1. Shown are the mean values of three biological replicates. Error bars show the standard deviations between these replicates.

Figure 6: Replacement of Tr- cbh1 with Ct- cbh1 gene in T. reesei increased the production of cellulases. (A) Schematic diagram of the construction of recombinant T. reesei strain Tr-Ct cbh1 . FgDNA-P indicates the fragment in the parental strain. FvDNA indicates the fragment from plasmid vector constructed for generation of the indicated recombinant strain. FgDNA-T indicates DNA fragment in the generated transformant strain. (B) Total secreted protein levels of three independent C pyr4 and Tr-Ct cbh1 transformants in the supernatant of 120-h Avicel cultures after transferring from glucose cultures. Shown are the mean values of three technical replicates. Error bars show the standard deviations between these replicates. (C) SDS-PAGE analysis of the extracellular proteins of C pyr4 and Tr-Ct cbh1 strains. The protein bands highlighted with the red frame line indicates proteins in Tr-Ct cbh1 strains were less than that in C pyr4 strains. (D) Filter paper activities and CBH activities of C pyr4 and Tr-Ct cbh1 strains. The samples used in (C,D) were as same as in (B) . Error bars in (D) show the standard deviations between the biological replicates shown in (C) . The significance of differences between C pyr4 and Tr-Ct cbh1 strains was based on t -test analysis by the FDR approach. Asterisks indicate significant differences (P 0.001). ns, not significant.

Figure 7: The activities of other cellulases were also increased in Tr-Ct cbh1 strains. (A) HPAEC-PAD analysis of glucose and cellobiose content in the solution of filter paper activity assay (FPA). nC, nano counts, indicating the intensity (abundance) of the signal for each peak. Glucose and cellobiose indicate 100 M glucose and cellobiose standard sample, respectively. Tr-Ct cbh1 and C pyr4 indicate sample taken from the solution of FPA assay with the supernatants of C pyr4 and Tr-Ct cbh1 strains under Avicel cultures, respectively. (B) Measurements of the endoglucanase activities and -glucosidase activities of C pyr4 and Tr-C tcbh1 strains in the supernatant of 120-h MM+2% Avicel cultures after transferring from 48 h of MM+2% glucose cultures. The significance of differences between C pyr4 and Tr-Ct cbh1 strains was based on t -test analysis by the FDR approach. Asterisks indicate significant differences (P 0.001). ns, not significant.

Figure 8: Increased CBH1 activities in T. reesei enhanced the induction and expression of the major cellulases genes. Relative quantification of mRNA levels of eg1 (A) , eg2 (B) , cbh2 (C) , and bgl1 (D) genes of C pyr4 and Tr-Ct cbh1 strains by qRT-PCR. RNA samples were obtained from the culture of MM+2% Avicel at the indicated time point after a shift of 48 h of MM+2% glucose cultures. The significance of differences between C pyr4 and Tr-Ct cbh1 strains was based on t -test analysis by the FDR approach. Asterisks indicate significant differences (P 0.05; P 0.001). ns, not significant.

Figure 9: The higher production of cellulases in Tr-Ct cbh1 strains was caused by increased expression level of xyr1. Relative quantification of mRNA levels of xyr1 (A) , cre1 (B) , ace1 (C) , ace2 (D) , and ace3 (E) genes of C pyr4 and Tr-Ct cbh1 strains by qRT-PCR. RNA samples were obtained from 12, 24, and 48 h of cultures of MM+2% Avicel after a shift of 48 h of MM+2% glucose cultures. The significance of differences between C pyr4 and Tr-Ct cbh1 strains was based on t -test analysis by the FDR approach. Asterisks indicate significant differences (P 0.05; P 0.01; P 0.001). ns,

not significant.

Figure 10: Graphic summary of this study. (A) Compared to T. reesei , C. thermophilum degrades celluloses more efficiently with lower production of cellulases. (B) The proposed model of the mechanism for higher production of cellulases in Tr-C tcbh1 strains. Replacement of Tr- cbh1 with Ct- cbh1 gene in T. reesei first increased the CBH1 activity, which could cause instantaneous accumulation of cellobiose. This extra cellobiose might function as a possible inducer to trigger more induction and expression of cellulases genes.

Mentioned biomedical entities:

Genes: CBH, CBH1, FPA, MA, MUC, Su, ace1, ace2, actin, bgl1, cbh1, cbh2, cel1b, cellulase, cre1, eg1, eg1, pyr4, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: C. thermophilum

Tissues: N/A

Pathways: N/A

Query Result 6/25

Influence of cis Element Arrangement on Promoter Strength in Trichoderma reesei

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Applied and Environmental Microbiology

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Keywords Frequency (TF): cellulase: 0.001; reesei: 0.002; cbh1: 0.022; xyr1: 0.002

Abstract:

Trichoderma reesei can produce up to 100 g/liter of extracellular proteins. The major and industrially relevant products are cellobiohydrolase I (CBHI) and the hemicellulase XYNI. The genes encoding both enzymes are transcriptionally activated by the regulatory protein Xyr1. The first 850 nucleotides of the cbh1 promoter contain 14 Xyr1-binding sites (XBS), and 8 XBS are present in the xyn1 promoter. Some of these XBS are arranged in tandem and others as inverted repeats. One such cis element, an inverted repeat, plays a crucial role in the inducibility of the xyn1 promoter. We investigated the impact of the properties of such cis elements by shuffling them by insertion, exchange, deletion, and rearrangement of cis elements in both the cbh1 and xyn1 promoter. A promoter-reporter assay using the Aspergillus niger, bgoxA gene allowed us to measure changes in the promoter strength and inducibility. Most strikingly, we found that an inverted repeat of XBS causes an important increase in cbh1 promoter strength and allows induction by xylan or wheat straw. Furthermore, evidence is provided that the distances of cis elements to the transcription start site have important influence on promoter activity. Our results suggest that the arrangement and distances of cis elements have large impacts on the strength of the cbh1 promoter, whereas the sheer number of XBS has only secondary importance. Ultimately, the biotechnologically important cbh1 promoter can be improved by cis element rearrangement.

Summary:

INTRODUCTION: Xyr1-binding sites (XBS) are a group of XBS that are found in the promoters of the cbh1and xyn1genes. XYR1 is a protein that is regulated by the XBP1 gene. Currently, the main research efforts focus on the transcriptional regulation of cellulase- and hemicellulases-encoding genes. Nonetheless, the impact of the arrangement ofciselements on promoter strength has hardly been investigated so far. Therefore, during this study, we investigated the role ofcisselements composed of xBS in the promors.

RESULTS: Theciselement IR increases promoter strength. We constructed expression cassettes bearing the deletion of the respectivecisselement in each promoter or a motif exchange where X was exchanged for IR in thecbh1promoter and termedcislelement X-IR and IR was exchange for X in thexyn1promotor and termedescribedciselelement Y-IR. The modified and native promoters were fused to theAspergillus nigergoxAgene, which is an established reporter system inTrichoderma species and subsequently transformed intoT. reeseiQM6atmus53pyr4. Using the reestablishment of pyr4as a marker, we investigated these strains by a carbon source replacement experiment.

DISCUSSION: In the past, relevant regions and motifs for the regulation of the xyn1 gene were identified. In this study, we present results demonstrating that the previously identified cis element IR, an inverted repeat of XBS, conveys transcriptional induction on xilan and wheat straw. We studied the role of this cs element by deleting only this specific motif and not by classic deletion analysis of the promoter. Phenotypic analysis was initially performed by carbon source replacement experiments because growth effects can be excluded and this allowed us to focus on the impact of a single carbon source on the promoteer activity.

Figures:

Figure 1: Schematic representation of the cbh1 and xyn1 promoters. Green arrows indicate XBS and their orientations, indicates the TATA signal, and ATG represents the start codon of the cbh1 or xyn1 gene. The blue box indicates the cis element Z (direct repeat of three XBS), the orange box indicates the cis element Y (an inverted repeat of XBS), the purple box indicates the cis element X (direct repeat of three XBS), and the brown box indicates the cis element IR (inverted repeat of XBS). Numbers on top indicate the distance from the ATG in base pairs.

Figure 2: Promoter schemes and GoxA activities of recombinant T. reesei strains after a carbon source replacement experiment. (A) Green arrows indicate XBS and their orientations, indicates the TATA signal, and ATG represents the start codon of the reporter gene goxA . Blue boxes indicate cis element Z, orange boxes indicate cis element Y, purple boxes indicate cis element X, and brown boxes indicate cis element IR. Numbers on top indicate the distance from the ATG in base pairs. (B) Indicated strains were pregrown, and then mycelia were transferred to MA medium without any carbon source (gray bars) or with 1% glycerol (blue bars), 1% d -glucose (red bars), 0.5 mM d -xylose (purple bars), 1% d -xylose (green bars), or 1.5 mM sophorose (orange bars), incubated for 8 h, and harvested, and GoxA activities were measured in the supernatants. GoxA activity is given in relation to biomass (dry weight). Symbols indicate results below quantification level () and results below the detection limit (). The mean values shown are calculated from the results of biological duplicates and technical triplicates. Error bars indicate standard deviations.

Figure 3: GoxA activities of strains bearing modified cbh1 promoters after direct cultivation. (A) Green arrows indicate XBS and their orientations, indicates the TATA signal, and ATG represents the start codon of the reporter gene goxA . Blue boxes indicate cis element Z, orange boxes indicate

cis element Y, purple boxes indicate cis element X, and brown boxes indicate cis element IR. Numbers on top indicate the distance from the ATG in base pairs. (B) Indicated strains were grown on MA medium with 1% glycerol (dark blue bars), lactose (red bars), xylan (green bars), carboxymethylcellulose (purple bars), or pretreated wheat straw (light blue bars). GoxA activity is given in relation to biomass (NaOH-soluble protein). The mean values shown are calculated from the results of biological duplicates and technical triplicates. Error bars indicate standard deviations.

Figure 4: GoxA activities of strains bearing modified cbh1 promoters after direct cultivation. (A) Green arrows indicate XBS and their orientations, indicates the TATA signal, and ATG represents the start codon of the reporter gene goxA . Blue boxes indicate cis element Z, orange boxes indicate cis element Y, purple boxes indicate cis element X, and brown boxes indicate cis element IR. Numbers on top indicate the distance from the ATG in base pairs. (B) Indicated strains were grown on MA medium with 1% glycerol (dark blue bars), lactose (red bars), xylan (green bars), carboxymethylcellulose (purple bars), or pretreated wheat straw (light blue bars). GoxA activity is given in relation to biomass (NaOH-soluble protein). The mean values shown are calculated from the results of biological duplicates and technical triplicates. Error bars indicate standard deviations.

Figure 5: GoxA activities of strains bearing modified cbh1 promoters after direct cultivation. (A) Green arrows indicate XBS and their orientations, indicates the TATA signal, and ATG represents the start codon of the reporter gene goxA . Blue boxes indicate cis element Z, orange boxes indicate cis element Y, purple boxes indicate cis element X, and brown boxes indicate cis element IR. Numbers on top indicate the distance from the ATG in base pairs. (B) Indicated strains were grown on MA medium with 1% glycerol (dark blue bars), lactose (red bars), xylan (green bars), carboxymethylcellulose (purple bars), or pretreated wheat straw (light blue bars). GoxA activity is given in relation to biomass (NaOH-soluble protein). The mean values shown are calculated from the results of biological duplicates and technical triplicates. Error bars indicate standard deviations.

Figure 6: GoxA activities of strains bearing combinations of modified cbh1 promoter fragments after direct cultivation. Thirty-five recombinant strains bearing different combinations of 3 and 5 fragments of the cbh1 promoter (compare Table 3) were grown on MA medium with 1% xylan (A), lactose (42 h cultivation time) (B), or pretreated wheat straw (C). The top value in each table cell gives the GoxA activity relative to that in strain PcWT. The bottom value in each table cell gives the absolute GoxA activity in relation to biomass, i.e., U/mg NaOH-soluble protein (A, C) or U/mg dry weight (B). Color key: the GoxA activity values of PcWT are highlighted in yellow; blue indicates lower and green indicates higher GoxA activities. The mean values shown are calculated from the results of technical triplicates. Error bars indicate standard deviations.

Mentioned biomedical entities:
Genes: AG, Ace1, Ace2, B, CA, Cre1, Gal4, GoxA, HRP, MA, MO, MP, TetR, WI, Xyr1, cbh1, cellulase, clb2, cyc, goxA, gpd, leu3, peroxidase, pyr4, xyn1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, protoplast

Organisms: Escherichia coli

Tissues: capillary

Pathways: N/A

Query Result 7/25

Trichoderma reesei XYR1 activates cellulase gene expression via interaction with the Mediator subunit TrGAL11 to recruit RNA polymerase II

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Keywords Frequency (TF): cellulase: 0.009; reesei: 0.004; cbh1: 0.003; xyr1: 0.013

Abstract:

The ascomycete Trichoderma reesei is a highly prolific cellulase producer. While XYR1 (Xylanase regulator 1) has been firmly established to be the master activator of cellulase gene expression in T. reesei, its precise transcriptional activation mechanism remains poorly understood. In the present study, TrGAL11, a component of the Mediator tail module, was identified as a putative interacting partner of XYR1. Deletion of Trgal11 markedly impaired the induced expression of most (hemi)cellulase genes, but not that of the major -glucosidase encoding genes. This differential involvement of TrGAL11 in the full induction of cellulase genes was reflected by the RNA polymerase II (Pol II) recruitment on their core promoters, indicating that TrGAL11 was required for the efficient transcriptional initiation of the majority of cellulase genes. In addition, we found that TrGAL11 recruitment to cellulase gene promoters largely occurred in an XYR1-dependent manner. Although xyr1 expression was significantly tuned down without TrGAL11, the binding of XYR1 to cellulase gene promoters did not entail TrGAL11. These results indicate that TrGAL11 represents a direct in vivo target of XYR1 and may play a critical role in contributing to Mediator and the following RNA Pol II recruitment to ensure the induced cellulase gene expression.

Summary:

Introduction: In eukaryotes, RNA polymerase II (Pol II) transcribes all protein-coding genes whose initiation is dependent on a set of general transcription factors (GTFs), which recognize the core promoter and facilitate transcription initiation from the correct start site. While various additional factors contribute to the regulation of Pol II activity at the promoter, the multisubunit Mediator complex has been shown to be critical for expression of most, if not all, Pol 2 transcripts.

Results: Identification of a T.reeseiGAL11/MED15 homolog that interacts with XYR1in vitro. (A) Schematic diagram of domain of TrGAL11. The KIX domain and glutamine rich region (Q-rich) of each protein are labeled as indicated. (B) Yeast two-hybrid analyses of interactions between Xyr1 activation domain (XYr1 AD) and full length or the KIx domain of TrGAL1in vivo.

Discussion: In this study, we showed that the major transactivator XYR1 directly targets the Mediator tail module subunit TrGAL11 to initiate (hemi)cellulase gene transcription in T. reesei. Tail module sub units including TrGal11 and TrMED5 were demonstrated to be differentially involved in the cellulasese gene expression. In addition, TrGaL11 and thus RNA Pol II binding to cellulasse gene promoters specifically relied on the expression of XyR1, whereas the steady state XYA binding to its target regulatory sequences was subject to a regulation imposed by TrGALA11 recruitment.

Figures:

Figure 1: TrGAL11 interacts with XYR1 in vitro .

Figure 2: Trgal11 disruption had little effect on T . reesei growth on plates or in liquid medium but reduced its conidia and pigment formation.

Figure 3: Deletion of Trgal11 compromised the fully induced production of cellobiohydrolase and endoglucanase but not of -glucosidase.

Figure 4: Deletion of Trgal11 resulted in a significant decrease in the transcription of cellobiohydrolase and endoglucanase but not -glucosidase genes.

Figure 5: Overexpression of xyr1 failed to restore the fully induced expression of cellulase genes in the Trgal11 deletion mutant.

Figure 6: TrGAL11 was recruited to cellulase gene promoters in an XYR1-dependent manner.

Figure 7: Trgal11 deletion resulted in a significantly higher XYR1 occupancy on cellulase gene promoters.

Figure 8: TrGAL11 plays a critical role in RNA Pol II recruitment on the core-promoter of cbh / eg genes but not of bgl genes.

Figure 9: A schematic model of how Mediator is recruited by XYR1 to participate in activating cellulase gene expression in T . reesei .

Mentioned biomedical entities:

Genes: B, Blue, CKM, FL, FPA, GAL4, GCN4, GST, HSF1, IP, MED14, MED16, MED2, MED3, MEL1, PIC, Rpb1, SD, TFIIH, UPS, XYR1, actin, bgl1, bgl2, cbh, cbh1, cbh2, cellulase, eg1, p53, pyr4, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: E. coli, human

Tissues: Tail

Pathways: N/A

Query Result 8/25

Regulation of gene expression by the action of a fungal lncRNA on a transactivator

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RNA Biology

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Abstract:

Long non-coding RNAs (lncRNAs) are crucial factors acting on regulatory processes in eukaryotes. Recently, for the first time in a filamentous fungus, the lncRNA HAX1 was characterized in the ascomycete Trichoderma reesei. In industry, this fungus is widely applied for the high-yield production of cellulases. The lncRNA HAX1 was reported to influence the expression of cellulase-encoding genes; interestingly, this effect is dependent on the presence of its most abundant length. Clearly, HAX1 acts in association with a set of well-described transcription factors to regulate gene expression. In this study, we attempted to elucidate the regulatory strategy of HAX1 and its interactions with the major transcriptional activator Xylanase regulator 1 (Xyr1). We demonstrated that HAX1 interferes with the negative feedback regulatory loop of Xyr1 in a sophisticated manner and thus ultimately has a positive effect on gene expression.

Summary:

Introduction: In recent decades, long non-coding RNAs (lncRNAs) have emerged as crucial players in regulatory processes in various eukaryotic organisms. Similar to other non-coded RNA species, lnnRNAs do not code for proteins but function directly as RNA. However, the group of lincRNA species is highly diverse and is distinguished from small functional ncRN species by their size (more than 200 nt), rather than by certain common characteristics ,.

Results: Fungal lncRNAHAX1and the transactivator Xyr1 can bind.As the recently discovered lincRNA HAX1, a physical interaction ofHAX1/XyR1 can be hypothesized. To test this possibility, RNA electrophoretic mobility shift assays were performed. To this end, the different versions ofHAx1varying in length (i.e.,HAX1262,HAX1299andHAX1428) were synthesizedin vitro, and XYr1 was expressed heterologously. The addition of increasing amounts of Xr1 toHAX1.

Discussion: In general, negative feedback regulation is a common strategy to ensure that a process stops when high amounts of the product are available. This regulation is essential to avoid unnecessary energetically demanding reactions and to maintain balance in complex metabolic pathways. However, despite the central role of Xyr1 in the regulation of PBDE expression, prior to this study, no evidence for possible auto-regulation of this transactivator had been reported. In this study we demonstrate that XYR1 regulates the expression of its own gene in a negative feedback loop via binding to the currently undescribed recognition site XRE based on bothin vitroandin vivoexperiments. Remarkably, the X RE was found to be strikingly different in terms of its properties as a XYA-binding motif compared to the YBS present in the promoters of xyr1.

Figures:

Figure 1: Analysis of the protein-RNA interaction of Xyr1 and HAX1 . RNA-EMSAs using either version of the lncRNA HAX1 , i.e., HAX1 262 (A), HAX1 299 (B) or HAX1 428 (C), and increasing amounts of Xyr1. An up to 4-fold molar excess of Xyr1 relative to 1g of in vitro synthesized HAX1 RNA was used.

Figure 2: Analyses of altered versions of the xyr1 promoter. (A) Schematic drawing of the xyr1 promoter (red bar). Previously described regulatory elements, such as Cre1-binding sites (yellow triangles), a Xyr1-binding site (green triangle) and a CAAT-box (blue box), as well as the newly identified 12 bp palindromic DNA motif XRE (pink box), are marked. The approximate positions of truncations are indicated by the numbers given on top (bp from ATG). (B) GoxA activities of recombinant T. reesei strains expressing the reporter gene goxA under the control of one of the truncated promoter versions (p804, p606, p497, p372) or the full-length version (pxyr1). The strains were pre-grown and replaced to minimal medium without a carbon source (blue bars) or to minimal medium with sophorose (orange bars). Means of biological replicates are derived from independently generated strains (n =2 for p804, p606, p497 and p372; n =3 for pxyr1). Error bars depict standard deviations; ANOVA followed by a post-hoc Tukey multiple comparison test (P 0.05) resulted in F (9;12)=11.684. Different letters denote significant differences among compared data. A detailed itemization of the P values is appended in Table S3. (C) GoxA activities of recombinant T. reesei strains cultivated as described above and expressing the reporter gene goxA under the control of the full-length promoter (pxyr1) or the full-length promoter lacking XRE (pXRE). Means of biological replicates are derived from independently generated strains (n =2 for pXRE; n =3 for pxyr1). Error bars depict standard deviations; t-tests resulted in t (3)=4.661, p 0.01 for the cultivation without carbon source and t (3)=3.326, p 0.05 for the cultivation on sophorose. Different letters denote significant differences among compared data. A detailed itemization of the P values is appended in Table S3.

Figure 3: Analysis of the binding of Xyr1 to XRE. (A) EMSA using 33.4 ng of the FAM-labelled probe containing the XRE alone (lane 1) or together with a 0.5-fold, 2-fold or 8-fold molar excess of Xyr1 relative to the probe (lanes 24). (B) Control EMSA using 33.4 ng of the FAM-labelled probe alone (lane

1) or with an 8-fold molar excess of Xyr1 (lane 2) or with an 8-fold molar excess of Xyr1 and increasing amounts (0.1-fold, 0.5-fold and 4-fold molar excess) of the unlabelled probe (Comp) (lanes 35) or with an 8-fold molar excess of BSA (lane 6).

Figure 4: Comparative analysis of the binding of Xyr1 to XRE and XBS. EMSAs using the FAM-labelled probe containing the XRE (Probe XRE) or the FAM-labelled probe containing XBS (Probe XBS) together with Xyr1, the unlabelled probe containing XBS (Comp XBS) or the unlabelled probe containing the XRE (Comp XRE). A total of 33.4 ng of the FAM-labelled probe alone (lanes 1) or together with a 4-fold molar excess of Xyr1 (lanes 2) or together with a 4-fold molar excess of Xyr1 and equimolar amounts of either of the unlabelled probes (lanes 3 and 4) were applied.

Figure 5: Analyses of the binding of Xyr1 and HAX1 to XRE and XBS. EMSAs using the FAM-labelled probe containing the XRE (A) or the FAM-labelled probe containing XBS (B) alone (lanes 1) or together with Xyr1 (lanes 2), or together with in vitro -synthesized CY5-labelled HAX1 428 (HAX1) (lanes 4) or together with both (lanes 3). As a control, HAX1 428 was also loaded alone (lanes 5). An 8-fold molar excess of Xyr1 and a 4-fold molar excess of RNA relative to 33.4 ng of the FAM-labelled probes were used. Fluorescence and image analyses were performed via FAM-scan (right images), CY5-scan (middle images) and ethidium bromide staining (left images).

Figure 6: Analyses of the binding of Xyr1 to XRE and XBS in the presence of HAX1 . EMSAs using the FAM-labelled probe containing the XRE (A) or the FAM-labelled probe containing XBS (B) alone (lanes 1) or together with different amounts of Xyr1 (lanes 24), or together with 8-fold molar excess of Xyr1 and increasing amounts of HAX1 262 (lanes 57) or together with 8-fold molar excess of Xyr1 and increasing amounts of HAX1 428 (lanes 810). (C) RNA-EMSA using 0.5g of in vitro -synthesized CY5-labelled HAX1 299 (HAX1 -CY5) alone (lane 1), or together with an 8-fold molar excess of Xyr1 (lane 2), or together with an 8-fold molar excess of Xyr1 and equimolar amounts of the FAM-labelled probe containing XBS (lane 3) or together with an 8-fold molar excess of Xyr1 and a 4-fold molar excess of the FAM-labelled probe containing XBS (lane 4). Fluorescence and image analyses were performed via FAM-scan (right image), CY5-scan (middle image) and ethidium bromide staining (left image).

Figure 7: Analyses of the impact of HAX1 on xyr1 expression. (A) Transcript levels of xyr1 in the presence and absence of hax1 . The hax1 deletion strain QM6a hax1 (green bars) and the parent strain QM6aloxP (blue bars) were pre-grown and transferred to minimal medium without carbon source or containing xylose (XO) or sophorose (S) for 3h. Transcript analysis was performed in technical triplicates. Transcript levels were normalized to act and sar1 , refer to cultivation without carbon source and are given in logarithmic scale (lg). Error bars indicate standard deviations; t-tests resulted in t (4)=7.948, p 0.01 for the cultivation on xylose and t (4)=4.395, p 0.05 for the cultivation on sophorose. Different letters denote significant differences among compared data. A detailed itemization of the P values is appended in Table S3. (B) Transcript levels of xyr1 in the presence or absence of hax1 824 overexpressed in trans . The hax1 overexpression strain OE hax1 (purple bars) and the reference strain QM6a tmus53 (blue bars) were pre-grown and transferred to minimal medium containing sophorose for 1h and 2h. Transcript analysis was performed in technical triplicates. Transcript levels were normalized to act and sar1 , refer to QM6a tmus53 cultivated for 1h and are given in logarithmic scale (lg). Error bars indicate standard deviations; t-tests resulted in t (4)=1.672, p 0.05 for the cultivation for 1h and t (4)=4.380, p 0.05 for the cultivation for 2h. Different letters denote significant differences among compared data. A detailed itemization of the P values is appended in Table S3. (C) GoxA activities of strains expressing the reporter gene goxA under the control of the full-length xyr1 promoter (pxyr1, blue bar) or the full-length xyr1 promoter lacking XRE (pXRE, orange bar) and a strain expressing the reporter gene goxA under the control of the full-length xyr1 promoter and overexpressing hax1 428 (pxyr1OE hax1 , red bar). The strains were pre-grown and transferred to minimal medium containing sophorose for 2h. GoxA assays were performed in technical triplicates. Error bars indicate standard deviations; t-tests resulted in t (4)=3.239, p 0.05 for the analysis of pxyr1OE hax1 versus pxyr1 and t (4)=0.292, p 0.05 for the analysis of pxyr1OE hax1 versus pXRE. Different letters denote significant differences among compared data. A detailed itemization of the P values is appended in Table S3.

Figure 8: Schematic illustration of the proposed regulatory model of HAX1 and Xyr1. Xyr1 (blue hexagon) can bind to two DNA motifs: XBS and XRE. Preferentially, Xyr1 binds to XBS at the promoter regions of its target genes (e.g., cbh1, cbh2, xyn1), thereby acting as a transcriptional activator (left section). When high amounts of Xyr1 are also available, the XRE on the xyr1 promoter gets bound, thereby resulting in a negative feedback regulation of xyr1 expression (right section). Via formation of an RNA-protein complex, HAX1 interferes with the negative feedback regulatory loop of Xyr1, rather than with its function as an activator. Thus, HAX1 finally has an enhancing effect on the expression of the Xyr1 target genes. Due to a higher affinity of Xyr1 for the longer versions of HAX1, HAX1 428 has the largest impact on PBDE expression, and HAX1 299 has a greater effect than the shortest version, HAX1 262.

Mentioned biomedical entities:
Genes: Cre1, GoxA, HAX1, MA, MO, MP, NEB, WI, XRE, Xyr1, bgl1, cbh1, cbh2, goxA, pyr4, sar1, tef1, xyn1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell

Organisms: Escherichia coli, human

Tissues: pulp

Pathways: N/A

Query Result 9/25

A homologous production system for Trichoderma reesei secreted proteins in a cellulase-free background

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Applied Microbiology and Biotechnology

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Keywords Hits: cellulase: 46, reesei: 15, cbh1: 4, xyr1: 16

Keywords Frequency (TF): cellulase: 0.019; reesei: 0.006; cbh1: 0.002; xyr1: 0.007

Abstract:

Recent demands for the production of biofuels from lignocellulose led to an increased interest in engineered cellulases from Trichoderma reesei or other fungal sources. While the methods to generate such mutant cellulases on DNA level are straightforward, there is often a bottleneck in their production since a correct posttranslational processing of these enzymes is needed to obtain highly active enzymes. Their production and subsequent enzymatic analysis in the homologous host T. reesei is, however, often disturbed by the concomitant production of other endogenous cellulases. As a useful alternative, we tested the production of cellulases in T. reesei in a genetic background where cellulase formation has been impaired by deletion of the major cellulase transcriptional activator gene xyr1. Three cellulase genes (cel7a, cel7b, and cel12a) were expressed under the promoter regions of the two highly expressed genes tef1 (encoding translation elongation factor 1-alpha) or cdna1 (encoding the hypothetical protein Trire2:110879). When cultivated on d-glucose as carbon source, the xyr1 strain secreted all three cellulases into the medium. Related to the introduced gene copy number, the cdna1 promoter appeared to be superior to the tef1 promoter. No signs of proteolysis were detected, and the individual cellulases could be assayed over a background essentially free of other cellulases. Hence this system can be used as a vehicle for rapid and high-throughput testing of cellulase muteins in a homologous background.

Summary:

Introduction: Fungal cellulases are attractive for enzymatic cellulose conversion as they are highly active and can be expressed at levels exceeding 100g per liter in fungal hosts such as Trichoderma reesei. Lignocellulose, composed mainly of cellulose, hemicellulose, and lignin, is the most abundant renewable carbon source on earth and therefore an attractive resource to use it for the production of different chemical building blocks or biofuels.

Results: Cellulase expression vector construction for the expression of cellulases in a xyr1strain inT. reesei. The expression of three cellullases was tested under both promoters. The cellullases were amplified by PCR and ligated downstream of thetef1promoter region. The cdna1 gene was cloned from the cDNA1 gene.

Discussion: XYR1 transcriptional regulator of cellulase and hemicellulases was not feasible, we show that the cloning of an expression vector based on the cdna1expression signals result in a reasonable high production of these enzymes ond-glucose as carbon source. In contrast to earlier reports where the coding sequence of the expression vector was not available, we present a novel approach to the characterization of cellulose degradation enzymes in T. reesei.

Figures:

Figure 1: Expression plasmids for cellulase production. Schematic presentation of the two cellulase expression plasmids used in this study. Both plasmids contain the hygromycin B expression cassette as fungal selection marker followed by either the tef1 or the cDNA1 promoter region and the coding and terminator region of the respective cellulase gene (e.g., the endoglucanase encoding gene cel7b)

Figure 2: Endoglucanase CEL7B, endoglucanase CEL12A, and cellobiohydrolase CEL7A production in the cellulase negative background of T. reesei xyr1 transformants. Expression profile of typical xyr1 transformants expressing CEL7B (a), CEL12A (b), or CEL7A (c) under the tef1 (t) or cdna1 (c) promoter regions as indicated. Left SDS-PAGE gel of the supernatant of selected transformants after Coomassie blue staining. The amount of supernatant loaded to each gel is indicated in parentheses. Right total extracellular protein concentration (in milligrams per liter) of selected transformants expressing the respective cellulase under the cDNA1 (filled diamond) or tef1 promoter region (filled square). xyr1 (empty triangle) is included as a negative control

Figure 3: Cellulase activities in the supernatants of the cel7b , cel7a , and cel12a transformants hydrolysis rates of MUC, MULAC, and CMC by culture supernatants of CEL7B (a), CEL12A (b), and CEL7A (c) producing strains respectively were assayed at each sampling point. Volumetric activities (dashed lines) are reflective of total active enzyme amount in the culture supernatant and specific activities (solid lines) are reflective of enzyme purity. Compared to the transformants the parental strain T. reesei xyr1 (empty triangle) has negligible activities towards MUC, MULAC, and CMC. The strains selected are identical to Fig. 2 and express the respective cellulase either under the cDNA1 (filled diamond) or tef1 promoter region (filled square). In the case of strain c-cel7b-6, extra d -glucose (10g/L) was added after 20h of incubation (empty diamond)

Mentioned biomedical entities:

Genes: AG, CEL12A, CEL3A, CEL6A, CEL7A, CEL7B, Cellulase, MUC, Tef1, XYR1, cbh1, cel12a, cellulase, egl3, tef1, xlnR, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: Escherichia coli

Tissues: N/A

Pathways: N/A

Query Result 10/25

Inducer-free cellulase production system based on the constitutive expression of mutated XYR1 and ACE3 in the industrial fungus Trichoderma, breesei

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Keywords Frequency (TF): cellulase: 0.019; reesei: 0.003; cbh1: 0.001; xyr1: 0.018

Abstract:

Trichoderma, bc toggle=yesreesei is a widely used host for producing cellulase and hemicellulase cocktails for lignocellulosic biomass degradation. Here, we report a genetic modification strategy for industrial T., bc toggle=yesreesei that enables enzyme production using simple glucose without inducers, such as cellulose, lactose and sophorose. Previously, the mutated XYR1V821F or XYR1A824V was known to induce xylanase and cellulase using only glucose as a carbon source, but its enzyme composition was biased toward xylanases, and its performance was insufficient to degrade lignocellulose efficiently. Therefore, we examined combinations of mutated XYR1V821F and constitutively expressed CRT1, BGLR, VIB1, ACE2, or ACE3, known as cellulase regulators and essential factors for cellulase expression to the T., bc toggle=yesreesei E1AB1 strain that has been highly mutagenized for improving enzyme productivity and expressing a -glucosidase for high enzyme performance. The results showed that expression of ACE3 to the mutated XYR1V821F expressing strain promoted cellulase expression. Furthermore, co-expression of these two transcription factors also resulted in increased productivity, with enzyme productivity 1.5-fold higher than with the conventional single expression of mutated XYR1V821F. Additionally, that productivity was 5.5-fold higher compared to productivity with an enhanced single expression of ACE3. Moreover, although the DNA-binding domain of ACE3 had been considered essential for inducer-free cellulase production, we found that ACE3 with a partially truncated DNA-binding domain was more effective in cellulase production when co-expressed with a mutated XYR1V821F. This study demonstrates that co-expression of the two transcription factors, the mutated XYR1V821F or XYR1A824V and ACE3, resulted in optimized enzyme composition and increased productivity.

Summary:

Introduction: Lignocellulosic biomass is the most abundant sustainable feedstock for biorefinery and biofuel production. Therefore, effectively utilizing lignocellulosesic biomass could assist in addressing the problem. Extracellular cellulases and hemicellulase for lignochellulusic biomass degradation are produced mainly by filamentous fungi. Especially the filamentous filamentous yeast Trichodermareesei(teleomorph Hypocreajecorna) is a well-known microorganism that produces large quantities of mixed extracellular cellulase and hemicellulase enzymes for the degradation of lignoscellulusc biomass.

Results: The constitutive expression of mutated XYR1V821Fand its candidate cooperative factors was combined with the disruption of repressors. The disruption of theace1andrce1repressorestrains was found to reduce protein productivity. The high-intensity band indicated by an open arrow was found in Fig.b. This protein was found as a glycoside hydrolase family 55 (GH55) protein (Trire2121746) by nano LCMS/MS analysis. This indicates that little cellulase production was observed in the three strains with glucose as the sole carbon source.

Discussion: Enhanced cellulase productivity using glucose as the sole carbon source was achieved by constitutive expression of mutated XYR1 V821F or XyR1 A824V and ACE3, especially ACE3 having a partially truncated DNA-binding domain. Previously, the deletion of these factors was reported to cause loss or reduction of celllulase expression,. These factors were also affected by enhanced expression of T. reesei.

Figures:

Figure 1: Effects of constitutive expression of V821F mutated XYR1 and disruption of repressors on glucose culture. (a) Extracellular protein from the cultivation of samples after 72, 96, and 120h. Trichoderma reesei strain E1AB1 was cultivated in shake flasks on an inducing medium containing 3% cellulose, and recombinant strains E1AB1 ace1 , E1AB1 rce1 , E1AB1-X (ace1 -P act1 - xyr1 V821F), and E1AB1-X rce1 were cultivated on a non-inducing medium containing 3% glucose. (b) SDS-PAGE analysis of the secreted proteins after 72h. T. reesei strain E1AB1 with 3% cellulose (lane 1) and 3% glucose (lane 2) cultures, and recombinant strains E1AB1 ace1 (lane 3), E1AB1 rce1 (lane 4), E1AB1-X (lane 5), and E1AB1-X rce1 (lane 6) with 3% glucose cultures. Open arrowhead indicates the confirmed in glucose culture (lanes 24). Closed arrowheads indicate clearly overexpressed proteins, likely corresponding to the main xylanase (BXL1, XYN1, XYN2, lanes 5 and 6). The gel was cropped in the 15250kDa range, and the original gel was shown in the Additional file Fig. S4. Error bars indicate standard deviations. Statistical significance was determined by a two-tailed unpaired Students t -test. p 0.01.

Figure 2: Co-expression of factors involved in cellulase transcription in the E1AB1-X strain. Trichoderma reesei strain E1AB1-X was co-expressed with various factors that relate to cellulase expression and cultivated in shake flasks on a non-inducing medium containing 3% glucose. (a) Extracellular protein from cultivation samples after 96h. (b) SDS-PAGE analysis of the secreted proteins after 72h in a glucose culture of T. reesei E1AB1-X (lane

1, ace1 -P act1 - xyr1 V821F , as control), E1AB1-XC (lane 2, ace1 -P act1 - xyr1 V821F and rce1 -P act1 - crt1), E1AB1-XB (lane 3, ace1 -P act1 - xyr1 V821F and rce1 -P act1 - bglr), E1AB1-XV (lane 4, ace1 -P act1 - xyr1 V821F and rce1 -P act1 - vib1), E1AB1-XA2 (lane 5, ace1 -P act1 - xyr1 V821F and rce1 -P act1 - ace2), and E1AB1-XA3 (lane 6, E1AB1-XA3, ace1 -P act1 - xyr1 V821F and rce1 -P act1 - ace3 (PT)). The gel was cropped in the 15250kDa range, and the original gel was shown in the Additional file Fig. S5. (c) Volumetric enzyme activity of the 72-h culture supernatant. One unit of activity is defined as the amount of enzyme that produced 1mol of p -nitrophenol per minute per mL of culture supernatant from the substrate at 50C. Bar graphs show the activity relative to that of E1AB1-X. Error bars indicate standard deviations. Statistical significance was determined by the two-tailed unpaired Students t -test. p 0.01.

Figure 3: Confirmation of the genetic combinations required for high cellulase production under inducer-free conditions. (a) Putative domain of XYR1 and ACE3 used in the study, amino acid mutations in XYR1, and the presumed variant of ACE3. Shaking flask cultivations were performed using 3% cellulose: Lane 1 and 3% glucose: Lane 29. The strains used in this figure were Lane 1 and 2: E1AB1, Lane 3: E1AB1-X (ace1 -P act1 - xyr1 V821F), Lane 4: E1AB1-A3 (rce1 -P act1 - ace3 (PT)), Lane 5: E1AB1-XA3 (ace1 -P act1 - xyr1 V821F and rce1 -P act1 - ace3 (PT)), Lane 6: E1AB1-XA3fl (ace1 -P act1 - xyr1 V821F and rce1 -P act1 - ace3 (FL)), Lane 7: E1AB1-XA3nhr (P act1 - xyr1 V821F and P act1 - ace3 (PT) by non-homologous recombination), Lane 8: E1AB1-X824A3nhr (P act1 - xyr1 A824V and P act1 - ace3 (PT) by non-homologous recombination) and Lane 9: E1AB1-XwtA3nhr (P act1 - xyr1 WT and P act1 - ace3 (PT) by non-homologous recombination).The carbon sources, strains, and genotypes used in this figure were organized in an Additional file, Table S3. (b) SDS-PAGE of secreted proteins after 72h cultivation (2.5g-protein/lane). The gel was cropped in the 15250kDa range, and the original gel was shown in the Additional file Fig. S6. (c) Extracellular protein concentration at 96h. Relative transcription levels of xyr1 (d) and ace3 (e). The real-time PCR was performed on the samples taken after 48h. The transcriptional levels of pgk1 and E1AB1 strain on glucose culture were measured for reference calculation and data normalization, respectively, and analyzed using the Ct method. Error bars indicate standard deviations. Statistical significance was determined by a two-tailed unpaired Students t -test. a indicated p0.01 for the test with Lane 5 (E1AB1-XA3) and b indicated p0.01 for the test with Lane 7 (E1AB1-XA3nhr).

Figure 4: Analysis of gene expression, enzyme activity, composition, and saccharification of microcrystalline cellulose. (a) Relative gene expression of the major cellulases and xylanases. The real-time PCR was performed on the 48h culture sample with glucose as a carbon source. The transcriptional levels of pgk1 and E1AB1-X strain RNA were measured for reference calculation and data normalization, respectively, and analyzed using the Ct method. (b) The volumetric activity of each enzyme in the culture supernatant obtained after 72h of cultivation with glucose as the carbon source. One unit of activity is defined as the amount of enzyme that produced 1mol of p-nitrophenol per minute per mL culture of the supernatant from the substrate at 50C. Bar graphs show relative activity to E1AB1-X. (c) Concentration of each enzyme component was calculated from the enzyme composition and total secreted protein concentration. The compositions were calculated from the band patterns following SDS-PAGE (see Additional file Fig. S2) derived from E1AB1 (culturing using cellulose: C and glucose: G), E1AB1-X (G), E1AB1-A3 (G), and E1AB1-XA3 (G) strains. Protein concentrations were determined from the 96-h value. (d) Microcrystalline cellulose saccharification using the same protein dosage (2.0mg protein/g cellulose). (e) Microcrystalline cellulose saccharification using the same volume of culture supernatant (0.87mL culture supernatant/g cellulose). Error bars indicate standard deviations. Statistical significance was determined by a two-tailed unpaired Students t -test. p 0.01, p 0.05.

Mentioned biomedical entities:

Genes: ACE1, ACE2, ACE3, ARE1, BGLR, BXL1, CBH, CBH1, CBH2, CIP2, CRE1, CRT1, CRZ1, CTF1, Cellulase, EG, EG1, Gal4, LAE1, MP, PAC1, RCE1, STR1, VEL1, VIB1, WT, XYN1, XYN2, XYR1, ace1, ace2, act1, cbh1, cbh2, cellulase, cre1, crt1, egl1, pgk1, pyr4, xyn1, xyn2, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: N/A

Tissues: tube

Pathways: N/A

Query Result 11/25

Tailoring the expression of Xyr1 leads to efficient production of lignocellulolytic enzymes in Trichoderma reesei for improved saccharification of corncob residues

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Biotechnology for Biofuels and Bioproducts

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Keywords Hits: cellulase: 96, reesei: 50, cbh1: 28, xyr1: 124

Keywords Frequency (TF): cellulase: 0.015; reesei: 0.008; cbh1: 0.004; xyr1: 0.019

Abstract:
The filamentous fungus Trichoderma reesei is extensively used for the industrial-scale cellulase production. It has been well known that the transcription factor Xyr1 plays an important role in the regulatory network controlling cellulase gene expression. However, the role of Xyr1 in the regulation of cellulase expression has not been comprehensively elucidated, which hinders further improvement of lignocellulolytic enzyme production.

Summary:
Background: The filamentous fungus Trichoderma reeseiis extensively used for the industrial-scale cellulase production. It has been well known that the transcription factor Xyr1 plays an important role in the regulatory network controlling cellulolytic enzyme gene expression. However, the role of Xr1 in the regulation of cellululase expression has not been comprehensively elucidated, which hinders further improvement of lignocellulolysis enzyme production.

Results: The xyr1 gene was overexpressed in T. reesei by cellulose-inducible promoters Pegl2, Pcbh1, and Pcdna1 promoter. The maximum transcript levels ofxyre1were found in the QCDX strain, followed by QCBX and QE2X. Strains were cultured at 30 and 200rpm in flasks using the CPM medium with 2 (w).

Conclusions: Our results underpin that the precise tailoring expression of xyr1is essential for highly efficient cellulase synthesis, which provide new insights into the role of Xyr 1 in regulating cellulolytic enzyme expression in T. reesei. Moreover, these results also provides a prospective strategy for strain improvement to enhance the lignocellulolysis enzyme production for use in biorefinery applications. 2016 Elsevier B.V.

Introduction: Lignocellulosic biomass is an abundantly available, inexpensive organic resource on the earth, which is basically composed of cellulose, hemicellulose, and lignin . Biological degradation of lignocellulocellulose biomass by lignochelolytic enzymes into fermentable sugars for the production of biofuels and other biochemicals has received extensive attention ,. However, the high cost of cellulase production, which was the major barrier for practical applications, limits the development of an efficient and economically viable lignoscellulusic feedstock based biorefinery . The filamentous fungus Trichoderma reeseiencodes a wide repertoire of cellobiohydrolases (CBHs), endoglucanases (EGs), and -glucosidase (exactly BGL1, EC 3.2.1.21) to degrade cellulose and other biomass components, which makes it be extensively employed in the industrial cellulolytic production .

Discussion: In T. reesei, genetic engineering of transcriptional regulators has been used as a powerful tool for enhancing cellulase production. Xyr1 is the key transcriptional activator of lignocellulolytic genes, whose overexpression can improve cellullase yields under both inducing and non-inducing conditions. However, the effect of xyr1/dosage regulation on cellulolase expression has not yet been elucidated. Herein, the expression of XYR1was tailored under promoters of different strengths and we found that the xYR1/overexpression with the Pegl2promoter resulted in a substantial improvement in cellululases production and saccharification efficiency. In previous studies, the promoter Ppdcwas chosen to drivexyR1expression for the construction of cellulesa hyperproducing strain ,,. In addition, the Promoter Ptcu1was also used to the expression , and the resultant strain could achieve full cellullasa expression . By roughly comparing the effects of the two promoter, we speculate that the overexpression ofxy1driven by the stronger promoter may have a more pronounced influence on cellulase production. However this does not imply that the maximization of cellulose production.

Conclusion: In the present study, we showed that the Pegl2-driven xyr1overexpression significantly improved production of lignocellulolytic enzymes in T. reesei. The total cellulase activity of QE2X was up to 8.4IU/mL under inducing condition. The glucose release from ACR by the enzyme mixture of Q E2X were improved by 57.2. To the best of our knowledge, this is the first report regarding the roles ofxyregulatory level on cellulolysis in T . eesii.

Figures:
Figure 1: Construction and validation of the xyr1 overexpression strains. A Schematic diagrams of the expression cassettes used for differential overexpression of the xyr1 gene. The overexpression of xyr1 was controlled by three different promoters (P egl2 , P cbh1, and P cdna1), respectively. B Transcript levels of xyr1 which was determined by RT-qPCR in QEB4 and three independent transformants of each xyr1 overexpression strain (QE2X, QCBX, and QCDX). Strains were cultivated at 30 and 200rpm in flasks using the CPM medium with 2% (w/v) Avicel as the sole carbon source for 72h. The transcript level was normalized to that of actin (t -test, p 0.01). Results are means of three biological replicates and error bars indicateSD

Figure 2: Detection of the cellulase secretion by T. reesei QEB4 and the xyr1 overexpression strains in different culture plates. The strains were assessed on the MCC plate (A), the CMC plate (C), and the CMC-esculin plate (E) to evaluate the secretion capacity of total cellulase, EGs and BGL, respectively. B The ratios between the diameter of the transparent zones and the colonies on MCC plate. D The hydrolytic halo diameters of the strains on CMC-Na plate. F The black zone diameters of the strains on CMC-esculin plate. p 0.05; p 0.01 (t -test). Results are means of three biological replicates and error bars indicateSD

Figure 3: Transcript analysis by RT-qPCR for genes encoding cellulases and regulators. Genes encoding cellulases (A) and transcription factors (B) were analyzed in the xyr1 overexpression strains with T. reesei QEB4 as the control. Strains were cultivated at 30 and 200rpm in flasks using the CPM medium with 2% (w/v) Avicel as the sole carbon source for 72h. The transcript levels of genes were normalized to that of actin (t -test, p 0.05, p 0.01, ns=not significant). Results are means of three biological replicates and error bars indicateSD

Figure 4: Transcript and CHART analysis of T. reesei QEB4 and the xyr1 overexpression strains. The genes of cbh1 (A), cbh2 (B), and the native xyr1 (C) were determined by RT-qPCR and CHART-PCR, respectively. The strains were cultured in CPM with 2% (w/v) Avicel for 48h. All data were normalized to the reference gene actin. The transcript levels are depicted on the x-axis and the chromatin accessibility indices (CAI) are plotted on the y-axis. Results are means of three biological replicates and error bars indicateSD

Figure 5: Transcript analysis of the genes encoding the ER-associated components by RT-qPCR. The UPR- (A) and ERAD- (B) related genes were analyzed in the xyr1 overexpression strains with T. reesei QEB4 as the control. Strains were cultivated at 30 and 200rpm in flasks using CPM with 2% (w/v) Avicel as the sole carbon source for 72h. The transcript levels of genes were normalized to that of actin (t -test, p 0.05, p 0.01, ns=not significant). Results are means of three biological replicates and error bars indicateSD

Figure 6: Cellulase activities and total extracellular proteins of T. reesei QEB4 and the xyr1 overexpression strains. A FPase activity (FPA), B cellobiohydrolase (CBH) activity, C endoglucanase (EG) activity, D -glucosidase (BGL) activity, and E the total extracellular protein of the fermentation supernatant from QEB4 and three recombinant strains cultured on 2% (w/v) Avicel for 7days. p 0.05; p 0.01, ns=not significant (t -test). Results are means of three biological replicates and error bars indicateSD

Figure 7: Saccharification of different pretreated corncob residues by T. reesei QEB4 and QE2X. Saccharification of acid-pretreated corncob residue (A) and delignified corncob residue (B) with the same FPA dosage. p 0.01, ns=not significant (t -test). Results are means of 3 biological replicates and error bars indicateSD

Figure 8: Transcript analysis for genes encoding accessory proteins of T. reesei QEB4 and QE2X by RT-qPCR. The transcript levels of genes were normalized to that of actin (t -test, p 0.05, p 0.01, ns=not significant). Results are means of three biological replicates and error bars indicateSD

Figure 9: Schematic summary showing the proposed model for the dosage regulation of xyr1 in T. reesei . A Compared to the strong xyr1 overexpression with the cbh1 / cdna1 promoter, the moderate xyr1 overexpression with the egl2 promoter can result in more open chromatin and stronger expression of cellulase gene, thus significantly enhancing cellulase production. B The lignocellulolytic enzyme production and saccharification efficiency were significantly improved in T. reesei QE2X, because of the high expression of cellulase and accessory proteins

Mentioned biomedical entities:
Genes: Ace1, Ace2, Ace3, B, BGL1, BglR, CBH, Cip1, Cip2, Cre1, Crz1, Ctf1, Dc, Ds, EG2, ER, FPA, Pac1, Rce1, Sec16, Swo1, Vib1, Xyr1, ace1, ace2, actin, bgl1, cbh1, cbh2, cellulase, cip1, cip2, cre1, ctf1, egl1, histone, hrd1, pdi1, ptrA, pyr4, swo1, vib1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 12/25

Development of a powerful synthetic hybrid promoter to improve the cellulase system of Trichoderma reesei for efficient saccharification of corncob residues

Wang, Yifan; Liu, Ruiyan; Liu, Hong; Li, Xihai; Shen, Linjing; Zhang, Weican; Song, Xin; Liu, Weifeng; Liu, Xiangmei; Zhong, Yaohua

Microbial Cell Factories

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Keywords Hits: cellulase: 44, reesei: 61, cbh1: 133, xyr1: 5

Keywords Frequency (TF): cellulase: 0.006; reesei: 0.008; cbh1: 0.018; xyr1: 0.001

Abstract:

The filamentous fungus Trichoderma reesei is a widely used workhorse for cellulase production in industry due to its prominent secretion capacity of extracellular cellulolytic enzymes. However, some key components are not always sufficient in this cellulase cocktail, making the conversion of cellulose-based biomass costly on the industrial scale. Development of strong and efficient promoters would enable cellulase cocktail to be optimized for bioconversion of biomass.

Summary:

Background: A synthetic hybrid promoter was constructed and applied to optimize the cellulolytic system of Trichoderma reeseifor efficient saccharification towards corncob residues. Firstly, a series of 5 truncated promoters in different lengths were established based on the strong constitutive promoter Pcdna1. The strongest promoter amongst them was Pccd1(640 to 1bp upstream of the translation initiation codon ATG), exhibiting a 1.4-fold higher activity than that of the nativecda1promoter. Meanwhile, the activation region (821 to 622bpupstream ofthe translation insertion codon and devoid of the Cre1-binding sites) of the strong inducible promoterPcbh1was cloned and identified to be an amplifier in initiating gene expression. Finally, this activation regions was fused to the strongest promoteer Pcb1d, generating the novel synthetic hybrid Promoter P.cc. This engineered promoter DROVE strong gene expression by displaying 1.6-fold.

Results: TEXT: Characterization of a series of 5 truncated promoters ofcdna1As a strong constitutive promoter inT. reesei, Pcdnam1has been successfully utilized to express cellulases using glucose as the carbon source . Here, the 928-bp promoter region upstream of ATG of the genecdm1was analyzed as shown in Fig.. The promoter Pcm1contains several corecis-acting elements, such as TATA box, CCAAT box and GC box, which are essential elements to initiate transcription or enhance the transcriptional level . To functionally validate thecdam1promoter, five 5 thuncated promors in different lengths were established based on Pcna1. Five 5 htuncted promoter, Ppcda1-1(928bp), Ppdnam1-2(794bp) Pppcnam1-3(640bp, 490bp and 247bp).

Conclusions: In this work, a synthetic hybrid promoter Pccwas created and confirmed to direct high-level expression of the essential cellulase components BGLA and EG2 to achieve the optimized cellulolytic enzyme system in T. reesei. Furthermore, when applied to the saccharification of differently pretreated corncob residues, this cellulolar enzyme system exhibited the prominent performance that displayed 2.3-fold higher saccharinefficacy than that of the parental strain. The chimeric promoter constructed here was the first hybrid promoteer applied in T . eesiand raised the possibility to generate powerful tunable promoters.

Introduction: The filamentous fungus Trichoderma reeseihas been widely used as an industrial workhorse for cellulase production due to its extraordinary ability to produce large amounts of extracellular cellulolytic enzymes. Among the cellullase enzymes secreted by this fungi, the cellobiohydrolase 1 (CBH1) accounts for more than 60 of total extracellular protein , thus thecbh1promoter (PcbH1) was generally used as a strong inducible promoter for gene expression .

Discussion: This study created a novel hybrid promoter Pccusing the synthetic approach in T. reesei, which enabled higher gene expression than that of the powerful promoters Pcbh1and Pcdna1. In T.reeei cost-effective cellulolytic enzyme system is needed to achieve efficient bioconversion of lignocellulosic biomass, and it could be obtained by modifying strains through improving cellulase expression levels and optimizing enzyme system ,. Promoter development is an essential method for strain modification and metabolic engineering . Thus, this study createda novel promoterPcc using the synthetic .

Figures:

Figure 1: Nucleotide sequences of the P cdna1 (a) and P cbh1 (b). The nucleotides ahead of A of translation initiation site (ATG) of cdna1 and cbh1 are numbered as 1. Known cis -acting elements in P cdna1 and mainly in the 869bp to 621bp region of P cbh1 are shown in different color. The descriptions of elements (TATA box, CAAT box, GC box, Xyr1-binding site, Ace2-binding site and Cre1-binding site) are shown in Additional file 1 : Table S1. An arrow above the sequence indicates the start point of different truncated fragments

Figure 2: Identification of the truncated fragments of P cdna1 fused with cbh1 as the reporter gene. a Strategy of targeted replacement of P cbh1 with the truncated fragments of P cdna1 fused with cbh1 . b , c CBH activity of the transformants PD1-1 (P cdna1-1), PD1-2 (P cdna1-2), PD1-3 (P

cdna1-3), PD1-4 (P cdna1-4), PD1-5 (P cdna1-5) and the parental strain QM53 (control) cultured in GMM for 5days (b) and AMM for 7days (c). Data indicates the mean of three independent experiments; error bars express the standard deviations. Different lowercase letters above the bars indicate significant differences at P0.05

Figure 3: Identification of P cbh1 activation region (AR) using egfp as the reporter gene. a Strategies of the construction of P cbh1-dc (P cbh1 devoid of Cre1-binding sites) and P cbh1-cr (core promoter of P cbh1) fused with egfp . RR: regulation region; CR: core region; AR: activation region. b , c Fluorescence relative ratio between the transformants PBD1 (P cbh1-dc-egfp), PBC1 (P cbh1-cr-egfp) and the parental strain QP4 cultured in GMM for 2days (b) and AMM for 3days (c). Data indicates the mean of three independent experiments; error bars express the standard deviations. Different lowercase letters above the bars indicate significant differences at P0.05

Figure 4: Construction of hybrid promoter P cc . a Strategies of the construction of P cc fused with the reporter gene egfp . b Fluorescence relative ratio between the transformants PB1 (P cbh1-egfp), PD1 (P cdna1-egfp), PCC and the parental strain QP4, respectively. Strains were cultured in AMM for 2days. c The fluorescence of transformant PCC (P cc-egfp) and the parental strain QP4 cultured with glucose for 24h and Avicel for 3days were observed under the fluorescence microscope. The results shown represent one of at least three independent experiments. Data indicates the mean of three independent experiments; error bars express the standard deviations. Different lowercase letters above the bars indicate significant differences at P0.05

Figure 5: Expression of bglA driven by P cc resulted in increase of BGL activity and FPA. Transformants expressing bglA driven by P cc and the parental strain QP4 cultured on glucose-esculin plates and CMC-esculin plates at 30 for 2days (a). SDS-PAGE analysis of the total extracellular cellulases of transformants and QP4 were shown after cultured in GMM for 5days (b) and CPM for 7days (c). BGLA bands were indicated by arrows. BGL activity (d) and FPA (e) of the transformants and QP4 were measured after cultured in GMM for 5days and CPM for 7days. Data indicates the mean of three independent experiments; error bars express the standard deviations. Different lowercase letters above the bars indicate significant differences at P0.05

Figure 6: Expression of eg2 driven by P cc resulted in increase of EG activity and FPA. SDS-PAGE analysis of the total extracellular cellulases of transformants and the parental strain QP4 were shown after cultured in GMM for 3days (a) and CPM for 7days (b). EG2 bands were indicated by arrows. EG activity (c) and FPA (d) of the transformants and QP4 were measured after cultured in GMM for 5days and CPM for 7days. Data indicates the mean of three independent experiments; error bars express the standard deviations. Different lowercase letters above the bars indicate significant differences at P0.05

Figure 7: Co-overexpression of bglA and eg2 driven by P cc in T. reesei resulted in increase of FPA. The transformants and the parental strain QP4 were cultured in CPM for 7days. FPA were measured at the end of cultivation. Data indicates the mean of three independent experiments; error bars express the standard deviations. Different lowercase letters above the bars indicate significant differences at P0.05

Figure 8: Saccharification of different pretreated corncob residues using the extracellular enzymes from QP4 and the transformants co-overexpressing bglA and eg2 driven by P cc . a Glucose released from acid-pretreated corncob residue at 48h. b Glucose released from delignified corncob residue at 48h. Data indicates the mean of three independent experiments; error bars express the standard deviations. Different lowercase letters above the bars indicate significant differences at P0.05

Mentioned biomedical entities:
Genes: AR, Ace1, Ace2, BGL1, BGLA, CBH, CBH1, Cre1, EG2, FPA, Hap5, PBC1, PBD1, PD1, Xyr1, bgl1, cbh1, cellulase, gpdA, hph, pdc, ptrA, pyrG

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 13/25

Systems biological approaches towards understanding cellulase production by Trichoderma reesei

Kubicek, Christian P.

Journal of Biotechnology

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Keywords Hits: cellulase: 78, reesei: 68, cbh1: 1, xyr1: 3

Keywords Frequency (TF): cellulase: 0.017; reesei: 0.015; cbh1: 0.000; xyr1: 0.001

Abstract:
First systems biology review on Trichoderma reesei. First omics comparison of Trichoderma reesei and other Trichoderma spp. The omics results offer novel understanding of biomass hydrolyzing enzymes. Routes for further research are illustrated.

Summary:
Introduction: The plant cell wall consists of the -(1,4)-linked glucose polymer cellulose, hemicellulose polysaccharides of varying composition, and lignin. The former two are formed by plants at an annual production rate of about 7.2 and 6 10 10 tons, respectively. They thus represent the largest reservoir of renewable carbon sources on earth which could be used for the production of biofuels and other biorefinery products, thereby replacing products derived from fossile carbon components.

Concluding remarks: As I have attempted to highlight above, the application of several -omics to T. reeseiand its formation of biomass degrading enzymes have led to a number of new and exciting insights into this process and its regulation, and many more findings will be published in the next future. In addition, tools for high throughput molecular genetics work with T.reeeihave been considerably expanded over the last years, and recipient strains with almost 100 integration into the homologous locus, a high throughpput system for gene disruption, methods for marker removement, several dominant selection markers, and mating type partners for sexual crossing are now available (for review see). It is therefore possible to tackle all important questions related to cellulase expression and secretion directly in this organism, and consequently there is no need to perform these investigations in other model fungi. The only disadvantage, which remains and which probably will not be rapidly overcome, is the comparatively small community of researchers working on the molecule biology of T.

Figures:
Figure 1: CAZys, indicated by GH-number, that are amplified in Trichoderma. Numbers on the vertical axes of the plots indicate number of gene paralogues. Abbreviations : J, T. reesei ; V, T. virens ; T, T. atroviride ; A, Aspergillus nidulans ; N, Aspergillus niger ; G, Gibberella zeae ; M, Magnaporthe grisea ; P, Podospora anserina ; B, Botryodiplodia fuckeliana ; S, Sclerotium sclerotiorum ; E, Neurospora crassa ; O, Rhizopus nigricans ; D, Batrachochytrium dendrobatidis . Data were compiled from Martinez et al. (2008) , Eastwood et al. (2011) , Goodwin et al. (2011) , and Kubicek et al. (2011, 2012) .

Figure 2: Genomic synteny of areas containing cellulase clusters, as determined by REEF. Synteny data were retrieved from the synteny browser at http://genome.jgi-psf.org/Trire2/Trire2.home.html , using the genome of T. atroviride as a comparison. Interrupted bars indicate lack of synteny. The different degrees of fillings of the bars (from black to grey) indicate occurrence of the respective areas on different scaffolds in T. atroviride . Numbers indicate the respective position on the scaffold in Mbps. Areas containing cellulase genes are indicated by red boxes.

Figure 3: Pedigree of H. jecorina mutant strains whose genome has been sequenced. The parent strain, whose genome sequence is already available is boxed; the cellulase negative mutants are located within the blue box.

Figure 4: Extracellular CAZymes identified by proteomic approaches by Jun et al., 2011 (A), Herpol-Gimbert et al., 2009 (B) and Adav et al., 2011 (C). Black boxes indicate that the protein was detected, numbers are the protein IDs in the T. reesei genome database. Note that the endoglucanase marked with an is actually a cellulose monooxygenase.

Mentioned biomedical entities:
Genes: ACE1, ACE2, ATF4, B, CBH1, CIP1, CPC1, CRE1, Cel6A, Cel7A, GH, GHs, H2A, H4, IRE1, LAE1, LaeA, MGST3, RIP, RUT, XYL1, XYR1, actin, cellulase, cpc1, cre1, gls2, histone, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: T. reesei, T. virens, mouse

Tissues: N/A

Pathways: N/A

Query Result 14/25

Interdependent recruitment of CYC8/TUP1 and the transcriptional activator XYR1 at target promoters is required for induced cellulase gene expression in Trichoderma reesei

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PLoS Genetics

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Keywords Frequency (TF): cellulase: 0.010; reesei: 0.005; cbh1: 0.000; xyr1: 0.012

Abstract:

Cellulase production in filamentous fungus Trichoderma reesei is highly responsive to various environmental cues involving multiple positive and negative regulators. XYR1 (Xylanase regulator 1) has been identified as the key transcriptional activator of cellulase gene expression in T. reesei. However, the precise mechanism by which XYR1 achieves transcriptional activation of cellulase genes is still not fully understood. Here, we identified the TrCYC8/TUP1 complex as a novel coactivator for XYR1 in T. reesei. CYC8/TUP1 is the first identified transcriptional corepressor complex mediating repression of diverse genes in Saccharomyces cerevisiae. Knockdown of Trcyc8 or Trtup1 resulted in markedly impaired cellulase gene expression in T. reesei. We found that TrCYC8/TUP1 was recruited to cellulase gene promoters upon cellulose induction and this recruitment is dependent on XYR1. We further observed that repressed Trtup1 or Trcyc8 expression caused a strong defect in XYR1 occupancy and loss of histone H4 at cellulase gene promoters. The defects in XYR1 binding and transcriptional activation of target genes in Trtup1 or Trcyc8 repressed cells could not be overcome by XYR1 overexpression. Our results reveal a novel coactivator function for TrCYC8/TUP1 at the level of activator binding, and suggest a mechanism in which interdependent recruitment of XYR1 and TrCYC8/TUP1 to cellulase gene promoters represents an important regulatory circuit in ensuring the induced cellulase gene expression. These findings thus contribute to unveiling the intricate regulatory mechanism underlying XYR1-mediated cellulase gene activation and also provide an important clue that will help further improve cellulase production by T. reesei.

Summary:

Introduction: CYC8/TUP1 corepressors are involved in the regulation of gene expression in eukaryotes. However, CY C8/ TUP1 has been implicated in gene activation in S.cerevisiae. Therefore, the role of CYc8/tup1 in gene regulation in S .cerevisiaceae is not well understood. Therefore , the CYP8/TUP1 corepression complex is a key component of the eutrophic repression system.

Results: The CYC8/TUP1 complex is a stable complex in T.reesei. The phylogenetic analysis revealed that the CYc8/ TUP1 sequence orthologs were widely distributed among filamentous ascomycete fungi including a few well-known cellulolytic Trichoderma, Penicillium, Fusarium, Neurospora, and Aspergillus strains but none of these putative CY C8/C1 homologs have been assessed regarding their functional involvement in plant cell wall degradation.

Discussion: The results of this study show that a TrCYC8/TUP1 complex exists in T. reesei, and is required for high-level activation of cellulase genesin vivo. Similar to functions in other filamentous fungi, repression ofTrcyc8 or Trtup1results in gross defects in vegetative growth and asexual spore production in T rese. Unlike the carbon catabolite repressor CRE1, the TrCYc8/ TUP1 Complex is critical for the induced cellullase gene expression. ChIP analyses reveal that TrTUP1/XYR1 are interdependent for efficient binding at cellululases gene promoters to ensure their induced expression.

Figures:

Figure 1: Evolutionarily conserved CYC8/TUP1 complex exists in T . reesei .

Figure 2: Knockdown of Trcyc8 and Trtup1 by conditional repression.

Figure 3: Knockdown of Trcyc8 or Trtup1 compromised T . reesei growth and conidiation on agar plates but not in liquid media.

Figure 4: Knockdown of Trcyc8 and Trtup1 abolished cellulase production.

Figure 5: Trcyc8 or Trtup1 repression resulted in a significant decrease in the transcription of cellulase gene and xyr1 transcription.

Figure 6: TrTUP1 was recruited to cellulase gene promoters upon induction in an XYR1-dependent manner.

Figure 7: XYR1 overexpression failed to rescue the defective cellulase production in P tcu1 - Trcyc8 KD or P tcu1 - Trtup1 .

Figure 8: Efficient binding of XYR1 to cellulase genes in vivo relied on TrTUP1.

Figure 9: The effect of Trcyc8 or Trtup1 repression on the association of histone H4 with cellulase gene promoters upon cellulose induction.

Figure 10: A working model for interdependent binding of TrCYC8/TUP1 and XYR1 on cellulase gene promoters to mediate their induced expression.

Mentioned biomedical entities:

Genes: CBH1, CEL7A, CRE1, CYC8, FZ, GCN4, H4, IP, LexA, MA, MEL1, MIG1, MO, PIC, SD, SspI, TPR, TUP1, WD40, XYR1, Xyr1, actin, arg1, arg4, bgl1, cbh1, cellulase, cre1, cyc1, fre2, histone, pyr4, tup1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: T. reesei

Tissues: tissue

Pathways: N/A

Query Result 15/25

A Mitogen-Activated Protein Kinase Tmk3 Participates in High Osmolarity Resistance, Cell Wall Integrity Maintenance and Cellulase Production Regulation in Trichoderma reesei, bFunction of a Hog1-Type MAPK (Tmk3) in T. reesei

Wang, Mingyu; Zhao, Qiushuang; Yang, Jinghua; Jiang, Baojie; Wang, Fangzhong; Liu, Kuimei; Fang, Xu; Harris, Steven

PLoS ONE

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Keywords Frequency (TF): cellulase: 0.010; reesei: 0.020; cbh1: 0.001; xyr1: 0.000

Abstract:

The mitogen-activated protein kinase (MAPK) pathways are important signal transduction pathways conserved in essentially all eukaryotes, but havent been subjected to functional studies in the most important cellulase-producing filamentous fungus Trichoderma reesei. Previous reports suggested the presence of three MAPKs in T. reesei: Tmk1, Tmk2, and Tmk3. By exploring the phenotypic features of T. reesei tmk3, we first showed elevated NaCl sensitivity and repressed transcription of genes involved in glycerol/trehalose biosynthesis under higher osmolarity, suggesting Tmk3 participates in high osmolarity resistance via derepression of genes involved in osmotic stabilizer biosynthesis. We also showed significant downregulation of genes encoding chitin synthases and a -1,3-glucan synthase, decreased chitin content, budded hyphal appearance typical to cell wall defective strains, and increased sensitivity to calcofluor white/Congo red in the tmk3 deficient strain, suggesting Tmk3 is involved in cell wall integrity maintenance in T. reesei. We further observed the decrease of cellulase transcription and production in T. reesei tmk3 during submerged cultivation, as well as the presence of MAPK phosphorylation sites on known transcription factors involved in cellulase regulation, suggesting Tmk3 is also involved in the regulation of cellulase production. Finally, the expression of cell wall integrity related genes, the expression of cellulase coding genes, cellulase production and biomass accumulation were compared between T. reesei tmk3 grown in solid state media and submerged media, showing a strong restoration effect in solid state media from defects resulted from tmk3 deletion. These results showed novel physiological processes that fungal Hog1-type MAPKs are involved in, and present the first experimental investigation of MAPK signaling pathways in T. reesei. Our observations on the restoration effect during solid state cultivation suggest that T. reesei is evolved to favor solid state growth, bringing up the proposal that the submerged condition normally used during investigations on fungal physiology might be misleading.

Summary:

Introduction: Cells sense their surrounding environments and react to external signals via signal transduction pathways. Among all known signal transducers, the mitogen-activated protein kinase (MAPK) pathway is ubiquitous in almost all eukaryotic species and is one of the most well characterized pathways. This pathway features a signal relay cascade in which three kinates are involved: the MAPK kin as kin (MAPKKK), the MAK kin as kin and MAPK.

Results and Discussion: Phylogenetic analysis of Tmk3 from T. reeseiand other previously characterized MAPKs showed Tm3 apparently cluster with Hog1-type MAPK from other species leading to the suggestion that TmK3likely encodes a Hog type MAPK. Sequence comparison between Tm1 from S. cerevisiaeand Hog3 fromT. euphorbiaeshowed a sequence identity of 66, further supporting this suggestion.

Conclusions: The Hog1-type MAPK Tmk3 in T. reesei is involved in high osmolarity resistance, cell wall integrity maintenance and cellulase production. It is homologous to Hog 1p from S. cerevisiae. Our results suggest that TmK3 functions in the cell wall Integrity signaling pathway similarly to Slt2-type MAKs. We further observed that deletion of Tmkk3leads to an apparent decrease in cellullase transcription, and suggested the involvement of Tk3in cellulose generation regulation by phosphorylating transcription factors.

Figures:

Figure 1: Growth of T. reesei parent strain and T. reesei tmk3 on plates.

Figure 2: Biomass accumulations in submerged media.

Figure 3: Test of sensitivity to NaCl, CR and CFW.

Figure 4: Response of transcriptional levels of glycerol and trehalose biosynthesis genes to elevated osmolarity.

Figure 5: Microscopic images of T. reesei parent strain and T. reesei tmk3 hyphae.

Figure 6: Transcriptional changes of chitin synthase- and -1,3-glucan synthase-coding genes.

Figure 7: Production of extracellular proteins, cellulases and hemicellulase in submerged media.

Figure 8: Transcriptional changes of cellulase- and hemicellulase-coding genes.

Figure 9: Biomass accumulations on solid state media.

Figure 10: Production of extracellular proteins, cellulases and hemicellulase in solid state media.

Mentioned biomedical entities:

Genes: ACEI , BXL1, CR, Cellulase, Chitinase, Cre1, E100, FPA, GPD1, Gna1, Gna3, Hog1, MAPK, MAPKKK, MO, MpkA, Slt2, Tmk1, Tmk2, Tmk3, Xyr1, bgl1, cbh1, cbh2, cellulase, chitinase, egl1, fus3, gpd1, pyr4

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: A. fumigatus, T. reesei

Tissues: tube

Pathways: N/A

Query Result 16/25

Structure-guided engineering of transcriptional activator XYR1 for inducer-free production of lignocellulolytic enzymes in Trichoderma reesei

Zhao, Qinqin; Yang, Zezheng; Xiao, Ziyang; Zhang, Zheng; Xing, Jing; Liang, Huiqi; Gao, Liwei; Zhao, Jian; Qu, Yinbo; Liu, Guodong

Synthetic and Systems Biotechnology

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Keywords Frequency (TF): cellulase: 0.005; reesei: 0.004; cbh1: 0.000; xyr1: 0.014

Abstract:

The filamentous fungus Trichoderma reesei is widely used for the production of lignocellulolytic enzymes in industry. XYR1 is the major transcriptional activator of cellulases and hemicellulases in T. reesei. However, rational engineering of XYR1 for improved lignocellulolytic enzymes production has been limited by the lack of structure information. Here, alanine 873 was identified as a new potential target for the engineering of XYR1 based on its structure predicted by AlphaFold2. The mutation of this residue to tyrosine enabled significantly enhanced production of xylanolytic enzymes in the medium with cellulose as the carbon source. Moreover, xylanase and cellulase production increased by 56.7- and 3.3-fold, respectively, when glucose was used as the sole carbon source. Under both conditions, the improvements of lignocellulolytic enzyme production were higher than those in the previously reported V821F mutant. With the enriched hemicellulases and cellulases, the crude enzymes secreted by the A873Y mutant strain produced 51% more glucose and 52% more xylose from pretreated corn stover than those of the parent strain. The results provide a novel strategy for engineering the lignocellulolytic enzyme-producing capacity of T. reesei, and would be helpful for understanding the molecular mechanisms of XYR1 regulation.

Summary:

Introduction: The production of highly-efficient and low-cost lignocellulolytic enzymes is important for bioconversion of plant cell wall resources. In Trichoderma reesei, the Zn2Cys6transcription factor XYR1 activates the expression of a major part of cellulases and xylanase genes ,. XyR1 isde novosynthesized in response to lignochellulasate-inducers (e.g. sophorose) , and activatesthe transcription of its target genes via recruiting the Mediator complex and RNA polymerase II . A single point mutation A824V was identified in XR1 in a xilanasese hyper-producingT. reistrain Iogen-M8, and this mutation was shown.

Results and discussion: The gene sequence of T. reesei xyr1was incorrectly annotated in JGI and UniProt databases. Actually, the second intron encodes peptide VSLASPSNQFQLQLSQPIFK, and the complete sequence ofT. eesiXYR1 contains 940 amino acid residues. The protein contains a Zn2Cys6DNA binding domain (92135) and a fungalTFMHR domain (fungal transcription factor regulatory middle homology region, 359843). This domain architecture is commonly observed for Gal4-like transcription factors in fungi .

Conclusions: A novel constitutively active mutant XYR1A873Ywas identified in T. reeseibased on the predicted structure of XyR1. The strain carrying this mutant showed significantly enhanced production of xylanolytic enzymes in cellulose medium, and expression of lignocellulolytic genes was activated under the inducer-free conditions. The crude enzymes produced by the mutant strain showed a better performance in the saccharification of corn fiber and pretreated corn stover compared with those of parent strain of the same protein loading. These findings provide an efficient strategy for improving the production level and performance of ligase system in T . eesi.

Figures:

Figure 1: Structural insights into the C-terminal domain of T. reesei XYR1. (A) The predicted structure of XYR1 in AlphaFold DB. The overall structure is colored according to per-residue confidence scores (pLDDT). Note that the amino acid sequence from 320 to 339 (VSLASPSNQFQLQLSQPIFK) was missed in the UniProt sequence (G0RLE8) used for structure prediction because of gene misannotation. In the close-up view, residues 821, 824 and 873 are shown in orange. (B) Sequence logos of the -helix regions (811833 and 868887) containing V821, A824 and A873.

Figure 2: Mutagenesis of A873 in T. reesei XYR1. (A) Schematic diagram of CRISPR/Cas9-aided mutagenesis. The 20-bp protospacer sequence is colored in orange. The PAM sequence is underlined. The donor templates contain homologous arms, mutated sequence (in red) and a selective PCR site (blue rectangle). (B) Extracellular protein concentrations of A873 mutation strains and parent strain QMP. The strains were cultivated in 2% (w/v) cellulose medium for 120h. Data represent meanSD from triplicate cultivations. The statistical significances of the difference between mutant strains and the parent strain are shown (, P 0.05; , P 0.01). (C) SDS-PAGE analysis of culture supernatants of equal volumes in cellulose medium at 120h. The supernatant of parent strain QMP was used as a control.

Figure 3: Lignocellulolytic enzyme production by XYR1 mutated strains in 2% (w/v) cellulose medium. (AE) Xylanase, -xylosidase, - l -arabinofuranosidase, filter paper enzyme, and cellobiohydrolase activities, respectively. Data represent meanSD from triplicate cultivations. (F) SDS-PAGE of culture supernatants of equal volumes at 120h.

Figure 4: Lignocellulolytic enzyme production by XYR1 mutated strains in 2% (w/v) glucose medium at 120h. (AD) Xylanase, -xylosidase, filter paper enzyme, and cellobiohydrolase activities, respectively. Data represent meanSD from triplicate cultivations. The statistical significances of the difference between mutant strains and the parent strain are shown (, P 0.05; , P 0.01; , P 0.001). (E) SDS-PAGE of culture supernatants of equal volumes.

Figure 5: The effects on transcript levels of lignocellulolytic enzyme encoding genes by xyr1 mutations. Strains were precultured in 1% (w/v) glycerol medium and thereafter transferred to media containing different carbon sources. The fold changes of transcript abundances relative to those in the parent strain are shown. Data represent meanSD from triplicate cultivations. (A) Transcript abundance changes in 1% (w/v) cellulose medium at 96h. (B, C) Transcript abundance changes in 2% (w/v) glucose medium at 18h and 90h, respectively.

Figure 6: Comparison of the structures of XYR1 wild type and A873Y mutant after MD. The wild-type structure is depicted in gray, while the mutants 765940 domain is depicted in a rainbow spectrum. The -helix offset directions are denoted by violet arrows, and labeled with numbers in red circles for reference. Key amino acid residues are presented in stick models. Interactions in the wild-type structure are illustrated with gray dashed lines, while those in the A873Y mutant are highlighted with yellow dashed lines. The disruption of the D802-R923 salt bond is indicated by a red dashed line. (AC) Three different perspectives of the conformational changes.

Figure 7: Releases of sugars from corn fiber by crude enzymes of T. reesei strains. The concentrations of total reducing sugar (A), xylose (B), glucose (C), cellobiose (D), arabinose (E) and galactose (F) are shown. Data represent meanSD from triplicate cultivations.

Figure 8: Releases of sugars from alkali pretreated corn stover by crude enzymes of T. reesei strains. The concentrations of glucose (A), xylose (B), arabinose (C) and cellobiose (D) at 72h are shown. Data represent meanSD from triplicate cultivations. The statistical significances of the difference between mutant strains and the parent strain QMP are shown (, P 0.001).

Mentioned biomedical entities:
Genes: AMA1, B, BXL1, Cas9, PAM, XYR1, cbh1, cellulase, pyr4, pyrG, sar1, xlnR, xyn2, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: Tobacco

Tissues: seed

Pathways: N/A

Query Result 17/25

The -importin KAP8 (Pse1/Kap121) is required for nuclear import of the cellulase transcriptional regulator XYR1, asexual sporulation and stress resistance in Trichoderma reesei

Ghassemi, Sara; Lichius, Alexander; Bidard, Frderique; Lemoine, Sophie; Rossignol, Marie-Nolle; Herold, Silvia; Seidl-Seiboth, Verena; Seiboth, Bernhard; Espeso, Eduardo A; Margeot, Antoine; Kubicek, Christian P

Molecular Microbiology

Year: 2015 PMID: 25626518 Citations: 14 Score: 54.51 DOI: 10.1111/mmi.12944

Keywords Hits: cellulase: 45, reesei: 87, cbh1: 4, xyr1: 55

Keywords Frequency (TF): cellulase: 0.007; reesei: 0.013; cbh1: 0.001; xyr1: 0.008

Abstract:

The ascomycete Trichoderma reesei is an industrial producer of cellulolytic and hemicellulolytic enzymes, and serves as a prime model for their genetic regulation. Most of its (hemi-)cellulolytic enzymes are obligatorily dependent on the transcriptional activator XYR1. Here, we investigated the nucleo-cytoplasmic shuttling mechanism that transports XYR1 across the nuclear pore complex. We identified 14 karyopherins in T.reesei, of which eight were predicted to be involved in nuclear import, and produced single gene-deletion mutants of all. We found KAP8, an ortholog of Aspergillus nidulansKapI, and Saccharomyces cerevisiaeKap121/Pse1, to be essential for nuclear recruitment of GFP-XYR1 and cellulase gene expression. Transformation with the native gene rescued this effect. Transcriptomic analyses of kap8 revealed that under cellulase-inducing conditions 42 CAZymes, including all cellulases and hemicellulases known to be under XYR1 control, were significantly down-regulated. kap8 strains were capable of forming fertile fruiting bodies but exhibited strongly reduced conidiation both in light and darkness, and showed enhanced sensitivity towards abiotic stress, including high osmotic pressure, low pH and high temperature. Together, these data underscore the significance of nuclear import of XYR1 in cellulase and hemicellulase gene regulation in T.reesei, and identify KAP8 as the major karyopherin required for this process.

Summary:

Introduction: Trichoderma reesei is a model system for the genetic and metabolic regulation of cellulase and hemicellulases. Cellula se gene expression is adaptive and inT.reeeiregulated by the action of at least four transcriptional activators (XYR1, ACE2, ACE3 and the HAP2/3/5 complex) and two repressors (ACE1 and the carbon catabolite repressionor CRE1) (for review, see Seiboth etal Amoreetal Kubicek,; Tani et al .,).

Results: Identification of the karyopherin- superfamily in T. reesei. We searched the T.reesian genome for potential orthologs of the 17 nuclear transporters previously identified in A. nidulans. We identified 14 proteins belonging to the superfamily. These proteins were assigned to the askap/KAP plus locus numbers 1 to 14. They comprised four potential exportins (KAP11A.nidullansKapKCrm1, KAP13 KapN, KAP5 KapECse1, KRP12 KapMLos1), nine -importins and the importin- homologue KAP1 KapASrp1. Additionalin silicosearches using Pfam domains related to this superfamily of proteins did not add more candidates to our predictions (data not shown).

Discussion: T.reesei-importin KAP8, an ortholog of S.cerevisiaePse1/Kap121 and A.nidulansKapan, is essential for the nuclear import of the transcriptional regulator for cellulase and hemicellulases gene expression XYR1. Pse1 was first reported in yeast as an auxiliary import receptor of ribosomal protein L25, since the defective import of L25 observed in thekap123 strain was reversed by the overexpression of Ps

Figures:

Figure 1: Expression of the T .reesei cellobiohydrolase cel7A gene during (A) growth on lactose for 16h (blue bars) and 36h (red bars) in nine -importin knockout strains. Values are given as relative gene expression, which is the ratio of expression of cel7A to that of the housekeeping gene tef1 , normalized to the same ratio obtained with the parental strain T .reesei QM9414 after 16h on lactose.B. Expression of cel7A in T .reesei parental strain (PS), kap8 and its complemented transformant kap8ct , and xyr1 after 24h preculture on 1% (w/v) glycerol and subsequent transfer to 1.4mM sophorose for 1h (grey bars) and 3h (white bars) respectively. Relative expression levels are defined as above, but the ratio obtained for the retransformant at t=0 was used for normalization. Error bars indicate the standard deviation from n 3 biological replicates.

Figure 2: Nuclear recruitment of GFP-XYR1 was not observed in the kap8 strain; however, it was restored in the kap8 complemented transformat (kap8 ct). In both cases, expression of gfp-xyr1 was under control of the native xyr1 promoter. On the left, fluorescent images are shown; on the right, the corresponding transmitted light images. Arrowheads indicate nuclei containing GFP-XYR1 and/or Hoechst dye. Scale bars, 10m.

Figure 3: Number of genes, grouped according to functional similarity, that were induced by sophorose at least fourfold (grey bars) and genes whose sophorose induction was significantly lower (5-fold) in the kap8 strain compared with the complemented transformant kap8ct (white bars). The grouping shown comprised 81.7% of all sophorose-inducible genes of strain kap8 ct as documented in Supporting Information TableS4. Gene groups and abbreviations: METABOLISM, genes involved in intracellular catabolism, anabolism and secondary metabolism; CAZY, all extracellular glycosyl hydrolases, carbohydrate esterases and accessory enzymes; ENZYMES REACTING WITH OXYGEN, oxidases, cytochrome P450- and FAD monooxygenases, dioxygenases; SSCRP, small, secreted, cysteine-rich proteins including hydrophobins and ceratoplatanins; SOLUTE TRANSPORT, members of the major facilitator superfamily; amino acid transport; ion transport; UNKNOWN PROTEINS, proteins that have orthologs in other

species, but for which no function has as yet been described; and ORPHAN PROTEINS, proteins that have orthologs, if at all, only in other Trichoderma species, and for which also no function is as yet known.

Figure 4: Growth of A .nidulans MAD2666 and mutant strains defective in selected kap genes (see Supporting Information TableS1) on D-glucose, the cellulose derivate CMC, xylan and D-xylose. Data from a single experiment are shown, but three replicates yielded consistent results. The order of strains and kap mutants for all carbon sources is shown in the sketch at the bottom of the figure.

Figure 5: Effect of kap8 knock out on asexual and sexual development in T .reesei :A. Phenotype of kap8 , the parental strain and the complemented transformant (kap8 ct) on PDA plates.B. Quantification of conidia harvested from these cultures. Vertical bars indicate standard deviations (n 3).C. fruiting body formation in confrontation with strain CBS999.79 (MAT 1-1).

Figure 6: Growth of T .reesei kap8 and the complemented transformant (kap8 ct) on PDA in the presence of various stress-inducing agents: from top to bottom 1.5, 10 and 20mM H 2 O 2 ; from top to bottom 0.5 and 2gml 1 fluconazole. Other concentrations/conditions are directly indicated. The plates shown are from a single experiments, but two further biological replicates yielded consistent results.

Figure 7: Inventory of gene groups that were not induced by sophorose but significantly down-regulated (at least 5-fold; P 0.05) in the kap8 strain compared with the retransformant. The gene groups encounter 152 from 163 down-regulated genes (for full description, see Supporting Information TableS6). Specification of gene groups is as described in the legend to Fig. 3 . New groups include HYDROLASES, extracellular lipases, esterases and amidases; SIGNAL TRANSDUCTION, involved in signal transduction pathways; and CELL WALL, proteins being components of the fungal cell wall, glycosyltransferases involved in their biosynthesis.

Mentioned biomedical entities:
Genes: ACE1, ACE2, ACE3, Aft1, AlcR, B, CRE1, ClrB, GCN20, Gal4, KAP1, KAP3, KAP6, KAP7, KAP8, Kap104, MAT1-1, MAT1-2, MFS, NirA, Nmd5, Pdr1, Pho4, Pse1, RUT, Ran, SLN1, Sln1, Ste12, Sxm1, Tel, XYR1, XlnR, Yap1, cbh1, cel7A, cellulase, kap1, pyr4, tef1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: Escherichia coli, spatula

Tissues: lens, pulp, tube

Pathways: N/A

Query Result 18/25

Engineering the Effector Domain of the Artificial Transcription Factor to Improve Cellulase Production by Trichoderma reesei

Meng, Qing-Shan; Zhang, Fei; Wang, Wei; Liu, Chen-Guang; Zhao, Xin-Qing; Bai, Feng-Wu

Frontiers in Bioengineering and Biotechnology

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Keywords Frequency (TF): cellulase: 0.016; reesei: 0.021; cbh1: 0.000; xyr1: 0.009

Abstract:

Filamentous fungal strains of Trichoderma reesei have been widely used for cellulase production, and great effort has been devoted to enhancing their cellulase titers for the economic biorefinery of lignocellulosic biomass. In our previous studies, artificial zinc finger proteins (AZFPs) with the Gal4 effector domain were used to enhance cellulase biosynthesis in T. reesei, and it is of great interest to modify the AZFPs to further improve cellulase production. In this study, the endogenous activation domain from the transcription activator Xyr1 was used to replace the activation domain of Gal4 of the AZFP to explore impact on cellulase production. The cellulase producer T. reesei TU-6 was used as a host strain, and the engineered strains containing the Xyr1 and the Gal4 activation domains were named as T. reesei QS2 and T. reesei QS1, respectively. Compared to T. reesei QS1, activities of filter paper and endoglucanases in crude cellulase produced by T. reesei QS2 increased 24.6 and 50.4%, respectively. Real-time qPCR analysis also revealed significant up-regulation of major genes encoding cellulase in T. reesei QS2. Furthermore, the biomass hydrolytic performance of the cellulase was evaluated, and 83.8 and 97.9% more glucose was released during the hydrolysis of pretreated corn stover using crude enzyme produced by T. reesei QS2, when compared to the hydrolysis with cellulase produced by T. reesei QS1 and the parent strain T. reesei TU-6. As a result, we proved that the effector domain in the AZFPs can be optimized to construct more effective artificial transcription factors for engineering T. reesei to improve its cellulase production.

Summary:

Introduction: Lignocellulosic biomass is abundantly available as a renewable resource, which is mainly composed of cellulose, hemicelluloses, and lignin. Degradation of the cellulose component by cellulase into glucose to produce biofuels and biobased chemicals has attracted extensive research attention. However, high production cost of cellulolase makes the bioconversion process too expensive for practical applications, which ultimately limits the utilization of lignocellulosesic biomass for biorefinery.

Results: The construction of the recombinant strainT. reeseiQS2 contains the novel artificial transcription factor AZFPM2-Xyr1AD. The AZFPs coding sequence is composed of four zinc fingers acting as a DBD, followed by a Gal4AD. Considering the heterologous origin of Gal4 AD, we were interested in whether cellulase production could be further improved when Gal4Ad was replaced by the endogenous Xy1AD and a relatively stronger promoter ofxyn3. Therefore, a new artificial transcription Factor AZFM2-XYr1Adwas constructed, and was constitutively expressed at thexyyn3locus inT.

Discussion: Yeast Gal4AD is a transcription activator for regulation of genes involved in galactose catabolism. Yest Gal4p is enriched in zinc finger DBDs, which can activate transcription by recruiting SAGA complex. Xyr1 is X-cluster protein, which is able to activate the expression of cellulase and hemicellulases gene expression inT. reesei.

Conclusion: We designed a new artificial transcription factor AZFP M2 -Xyr1 AD in T. reesei, and found that AZFPM2-Xyl1ADcould significantly improve the expression of major cellulase genes. Moreover, the cellullase produced by the T.reeei QS2 carrying AZPFM2-Xl1 ADwas proven to be highly effective in the saccharification of the pretreated lignocellulosic biomass. As a result, we proved that the endogenous effector domain Xyler1ADis more advantageous for construction of AZPP not only in improving cellulolytic production, but also in optimizing cellululases complex for biomass conversion.

Figures:

Figure 1: Construction of fusion transcription factor AZFP M2 -Xyr1 AD in T. reesei . (A) Schematic map of AZFP M2 -Xyr1 AD . (B) Schematic diagram of Southern bolt analysis. (C) Southern blot analysis of transformants QS1 and QS2 with AZFPs at xyn3 loci, respectively. Southern blot analysis of the genome digested with Nde I. A fragment of 3.7 kb is present in the parent strain, and the 5.7 and 7.1 kb bands are shown in the mutant strains QS1 and QS2, respectively. (D) Western blot analysis for AZFP M2 -Gal4 AD /Xyr1 AD in T. reesei QS1 and QS2, respectively. T. reesei TU-6 represent the parent strain.

Figure 2: Cellulase production by T. reesei QS2, QS1 and TU-6. (AE) The activities for FPase, CMCase, p NPCase, p NPGase and xylanase, respectively. (F) Total extracellular proteins content. The strains were cultured at 28C and 180 rpm in flasks using minimal medium supplemented with 2% cellulose and 2% wheat bran as carbon source. Error bar denotes the standard deviations (SD) of three replicates (p 0.05, p 0.01).

Figure 3: Transcription analysis of genes encoding major cellulolytic enzymes and regulators. Genes encoding (A) cellulolytic enzymes, (B) regulators, and (C) accessory proteins were analyzed for T. reesei QS1 and QS2 mutants with T. reesei TU-6 as the reference strain. Strains were cultured at 28C and 180 rpm in flasks using minimal medium supplemented with 2% cellulose and 2% wheat bran as a carbon source for 24 and 48 h, respectively. Expression levels of the reference gene tef1 were used as an endogenous control. The value is the mean of three biological replicates with SD as the error bars. FC represents fold change of the transcription levels with targeted genes detected in the mutants over that detected in the control.

Figure 4: Hydrolysis of alkali pretreated corn stover (APCS) and jerusalem artichoke stalk (APJAS) by cellulases produced by T. reesei QS2, QS1, and TU-6. Sugar release from (A) alkali pretreated corn stover (APCS) and (B) alkali pretreated jerusalem artichoke stalk (APJAS) was examined. The same protein dosage (30 mg/g substrate) was used for enzymatic hydrolysis experiments. Data are represented as the mean of triplicate with standard deviation (SD) as the error bars.

Mentioned biomedical entities:

Genes: AD, ADs, APCS, Ace1, Ace2, Ace3, Cel61A, Cellulase, Cip1, Cip2, Cre1, Ctf1, FZ, Fang, Gal4, Gcn5, MFS, Meng, Rce1, Swo1, VP16, Vib1, Xyr1, bgl1, cbh1, cbh2, cellulase, cip2, ctf1, egl1, pyr4, tef1, xyn2, xyn3, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 19/25

Modulating Transcriptional Regulation of Plant Biomass Degrading Enzyme Networks for Rational Design of Industrial Fungal Strains

Alazi, Ebru; Ram, Arthur F. J.

Frontiers in Bioengineering and Biotechnology

Year: 2018 PMID: 30320082 Citations: 15 Score: 49.83 DOI: 10.3389/fbioe.2018.00133

Keywords Hits: cellulase: 79, reesei: 31, cbh1: 2, xyr1: 41

Keywords Frequency (TF): cellulase: 0.012; reesei: 0.005; cbh1: 0.000; xyr1: 0.006

Abstract:
Filamentous fungi are the most important microorganisms for the industrial production of plant polysaccharide degrading enzymes due to their unique ability to secrete these proteins efficiently. These carbohydrate active enzymes (CAZymes) are utilized industrially for the hydrolysis of plant biomass for the subsequent production of biofuels and high-value biochemicals. The expression of the genes encoding plant biomass degrading enzymes is tightly controlled. Naturally, large amounts of CAZymes are produced and secreted only in the presence of the plant polysaccharide they specifically act on. The signal to produce is conveyed via so-called inducer molecules which are di- or mono-saccharides (or derivatives thereof) released from the specific plant polysaccharides. The presence of the inducer results in the activation of a substrate-specific transcription factor (TF), which is required not only for the controlled expression of the genes encoding the CAZymes, but often also for the regulation of the expression of the genes encoding sugar transporters and catabolic pathway enzymes needed to utilize the released monosaccharide. Over the years, several substrate-specific TFs involved in the degradation of cellulose, hemicellulose, pectin, starch and inulin have been identified in several fungal species and systems biology approaches have made it possible to uncover the enzyme networks controlled by these TFs. The requirement for specific inducers for TF activation and subsequently the expression of particular enzyme networks determines the choice of feedstock to produce enzyme cocktails for industrial use. It also results in batch-to-batch variation in the composition and amounts of enzymes due to variations in sugar composition and polysaccharide decorations of the feedstock which hampers the use of cheap feedstocks for constant quality of enzyme cocktails. It is therefore of industrial interest to produce specific enzyme cocktails constitutively and independently of inducers. In this review, we focus on the methods to modulate TF activities for inducer-independent production of CAZymes and highlight various approaches that are used to construct strains displaying constitutive expression of plant biomass degrading enzyme networks. These approaches and combinations thereof are also used to construct strains displaying increased expression of CAZymes under inducing conditions, and make it possible to design strains in which different enzyme mixtures are simultaneously produced independently of the carbon source.

Summary:
Introduction: Plant biomass is the most abundant renewable carbon source in the world and represents the major natural substrate for fungi (Kowalczyk et al). Plant biomass mainly consists of plant cell wall material, which contains the polysaccharides, i.e., cellulose, hemicellulose and pectin, lignin, and structural proteins (Loqu e. al). Cellulose, the most abundance plant cellwall polysoccharide, is a linear chain of glucose (KOLpak and Blackwell). Hemicelluloses are complex heteropolysaccharides with xylan- (linear chain of xilose), glucan- (linesar chain glucose), Glucomannan- (linesar chain glucos and mannose), or mannan- (linear chain mannose) backbones, with several different types of monomers attached to the backbone to give eg., arabinoxylan or xychlucan (Scheller and Ulvskov). Pectains are other complex polysacharides containing D-galacturonic acid (PGA) as the main sugar acid in their backbone. Polygalacturenic acid (PGA) is PGA and the most prevalent pectic substructure. Plant biomass also contains the plant cell storage polysaches starch, a linear (amylose) or branched (amielopectiin) polymer of glucose, and inulin, which are polymers of fructose residues with a terminal glucose residue (Gidley,; Ritsema and Smeekens). Plant biomass is the major renewable carbon substrate in the World and represents a major natural substratum for fungi (L.

Figures:

Mentioned biomedical entities:

Genes: AR, Ace1, AmyR, Ara1, AraR, ClrA, ClrB, Cre1, CreA, CreB, Fang, GA, GH, Gcn5, Gh1-1, Gh3-4, Lae1, ManR, QutA, QutR, Rce1, TF, XlnR, Xyr1, Yao, abfB, ace1, amyR, amylase, araR, bgl2, cbh1, ccg-1, cellulase, clrB, cre-1, cre1, creA, gpd, gpdA, manR, pdc, pdr-1, qutA, tef1, xlnR, xyn1, xyn2, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: A. nidulans, T. reesei

Tissues: N/A

Pathways: N/A

Query Result 20/25

Functional Characterization of Sugar Transporter CRT1 Reveals Differential Roles of Its C-Terminal Region in Sugar Transport and Cellulase Induction in Trichoderma reesei

Wang, Zhixing; Yang, Renfei; Lv, Wenhao; Zhang, Weixin; Meng, Xiangfeng; Liu, Weifeng; Atanasova, Lea

Microbiology Spectrum

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Keywords Frequency (TF): cellulase: 0.012; reesei: 0.003; cbh1: 0.001; xyr1: 0.008

Abstract:

The expression of cellulase genes in lignocellulose-degrading fungus Trichoderma reesei is induced by insoluble cellulose and its soluble derivatives. Membrane-localized transporter/transceptor proteins have been thought to be involved in nutrient uptake and/or sensing to initiate the subsequent signal transduction during cellulase gene induction. Crt1 is a sugar transporter proven to be essential for cellulase gene induction although the detailed mechanism of Crt1-triggered cellulase induction remains elusive. In this study, we focused on the C-terminus region of Crt1 which is predicted to exist as an unstructured cytoplasmic tail in T. reesei. Serial C-terminal truncation of Crt1 revealed that deleting the last half of the C-terminal region of Crt1 hardly affected its transporting activity or ability to mediate the induction of cellulase gene expression. In contrast, removal of the entire C-terminus region eliminated both activities. Of note, Crt1-C5, retaining only the first five amino acids of C-terminus, was found to be capable of transporting lactose but failed to restore cellulase gene induction in the crt1 strain. Analysis of the cellular localization of Crt1 showed that Crt1 existed both at the plasma membrane and at the periphery of the nucleus although the functional relevance is not clear at present. Finally, we showed that the cellulase production defect of crt1 was corrected by overexpressing Xyr1, indicating that Xyr1 is a potential regulatory target of the signaling cascade initiated from Crt1.

Summary:

INTRODUCTION: Lignocellulose, as the most abundant and cost-effective sustainable resource, exhibits tremendous potential in the production of biofuel and bio-based chemicals. However, due to the recalcitrant nature and complex structure of lignocellulose, there are inherent challenges in its bioconversion. Known for its high capacity to secrete cellulases, the filamentous fungus Trichoderma reesei has been used as the principal microorganism for industrial cellulosae production. While stringently controlled by catabolite repression, the expression of cellullase genes in T. reseis can be rapidly induced by either insoluble crystalline cellulose or soluble substrates, such as cellobiose, sophorose, and lactose). Regarding the induction of cellulases by insoluble substrate, it is generally acknowledged that the basal level of cellosales produced by the fungi act on the insoluble substrat to release and even transform soluble inducers to facilitate cellululase gene expression. However , the precise mechanism of how soluble inducers are perceived to initiate the signal transduction cascade leading to efficient cellulusa gene expression inT. reaesiis still enigmatic.

RESULTS: Subcellular localization of Crt1-GFP under cellulase induction condition. (A) Fluorescence microscopy analysis of Cr1-GFP after induction with 1 (wt/vol) Avicel in MA medium for 12 h. (B) Colocalization analysis of Crt1 and the NE-localized Sec61-GFp after inducing with 1 wt / vol Avicel for 12h. Scale bar: 10m. (C) Colocation analysis of the ne-localised Sec61-GF

DISCUSSION: The cellulose-degrading fungus T. reeseiis one of the most prominent cellulase producers in nature and is widely used as a workhorse for industrial cellullase production. Although the transcriptional regulation of cellulose gene expression has been intensively studied, the precise mechanism of how T. Reesi senses environmental nutrition cues to initiate intracellular signaling cascade remains largely unclear. Several fungal transporters have been shown to be involved in regulating celluluasae expression. Until now, Crt1 is considered to be the most important transporter in T.

Figures:

Figure 1: Subcellular localization of Crt1-GFP under cellulase induction condition. (A) Fluorescence microscopy analysis of Crt1-GFP after inducing with 1% (wt/vol) Avicel in MA medium for 12 h. Inset a: arrows indicated the plasma membrane; Inset b: arrows indicated the nuclear envelope. Scale bar: 10m. (B) Colocalization of Crt1-GFP and mCherry-Xyr1 after inducing with 1% (wt/vol) Avicel for 12 h. Scale bar: 10m. (C) Colocalization analysis of Crt1-GFP and the NE-localized Sec61-mCherry. The strain was cultured in MA medium with 1% (wt/vol) Avicel for 12 h and mycelia were collected for fluorescence analysis. Scale bar: 10m. The fluorescence was examined with a Nikon Eclipse 80i fluorescence microscope. The images shown are taken from one of at least two independent experiments.

Figure 2: Lactose transport activity analysis of wild type Crt1 and its C-terminal mutants. (A) Intracellular fluorescence analysis of wild type Crt1 and its different mutants in yeast cells. GFP was fused at the C-terminus of Crt1 and its mutants. The C-terminus sequence of each mutant was shown on the right. Images shown represent one of at least two independent experiments. Scale bar: 2m. (B) The growth (OD600 nm) of engineered S. cerevisiae W303 strains expressing wild-type Crt1 or its mutants in SC medium with 1% (wt/vol) lactose as the carbon source. S. cerevisiae W303 strains were equipped with the intracellular -galactosidase LAC4 from K. lactis before expressing Crt1 or its mutants. The lactose permease Lac12 from K. lactis

was used as the positive control (black square) and the empty vector as the negative control (gray dot). The results shown in this figure are the mean of three biological replicates.

Figure 3: Functional complementation test of Crt1 C-terminal mutants in the crt1 strain. Extracellular p NPC hydrolytic activities (A and C) and p NPG hydrolytic activities (B and D) of supernatants from the parental strain TU-6, Crt1 deletion strain crt1 and different complementation strains cultured in MA medium containing 1% (wt/vol) lactose (A and B) or Avicel (C and D) as the sole carbon source for indicated time periods were determined. The wild type Crt1 and its mutants were expressed under the P tcu1 promoter in their corresponding complementation strains. The values shown in this figure are the mean of three biological replicates. Error bars represent the standard deviation (SD) of these replicates. Statistical significances of extracellular activities relative to wild type TU-6 were determined using Students t test (n.s., P 0.05; , P 0.05; , P 0.01; , P 0.001).

Figure 4: Transcriptional analysis of major cellulase genes and xyr1 in TU-6, crt1 , and different complementary strains under lactose inducing condition. Relative transcription levels of main cellulase genes (cbh1 A, cbh2 B, and eg1 C) and xyr1 (D) were determined by quantitative RT-qPCR in a resting-cell system under 1% (wt/vol) lactose inducing condition. Values are the mean of three biological replicates and error bars represent the standard deviation (SD) of these replicates. The expression level of the actin gene was used as an endogenous control for all samples. Statistical significances of the target gene transcription relative to 0 h of each strain were determined using Students t test (n.s., P 0.05; , P 0.05; , P 0.01; , P 0.001).

Figure 5: The carbon catabolite repression on the cellulase gene transcription was not fully relieved by overexpression of crt1 . Transcriptional levels of cbh1 (A), eg1 (B), and crt1 (C) in TU-6, Re crt1 , and Re crt1- C5 strain were determined by quantitative RT-PCR at indicated time points in the resting-cell system using 1% lactose and 0.1% glucose as the carbon source. Values are the mean of three biological replicates and error bars represent the standard deviation (SD) of these replicates. The expression level of the actin gene was used as an endogenous control for all samples. Statistical significances of the target gene transcription relative to 0 h of each strain were determined using Students t test (n.s., P 0.05; , P 0.05; , P 0.01; , P 0.001).

Figure 6: Xyr1 is essential but not sufficient for the transcription of crt1 . The transcription levels of the endogenous xyr1 (A or D), cbh1 (B or E), and crt1 (C or F) in QM9414, xyr1 , and OE xyr1 strain were analyzed by quantitative RT-PCR at indicated time points. All strains were cultured in MA medium containing 1% (wt/vol) Avicel as the sole carbon source. Values are the mean of three biological replicates and error bars represent the standard deviation (SD) of these replicates. Statistical significances of the target gene transcription relative to wild type QM9414 were determined using Students t test (n.s., P 0.05; , P 0.05; , P 0.01; , P 0.001).

Figure 7: Overexpression of Xyr1 in crt1 bypassed its cellulase expression defect. Extracellular p NPC hydrolytic activities of supernatant from TU-6, crt1 , OE xyr1 , OE xyr1- crt1 strains cultured in MA medium containing 1% (wt/vol) glucose (A) or 1% (wt/vol) Avicel (B) were determined. Values are the mean of three biological replicates and error bars represent the standard deviation (SD) of these replicates. Statistical significances of extracellular activities relative to wild type TU-6 were determined using Students t test (n.s., P 0.05; , P 0.05; , P 0.01; , P 0.001).

Mentioned biomedical entities:

Genes: ADH1, AscI, B, CDT-2, CRT1, Cellulase, Crt1, EGFR, ER, FIG, GLUT1, HXT, MA, Mep2, Rgt2, SD, Sec61, SspI, Stp1, Xyr1, actin, cbh1, cbh2, cellulase, crt1, eg1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: Escherichia coli, K. lactis, T. reesei

Tissues: N/A

Pathways: N/A

Query Result 21/25

The protein methyltransferase TrSAM inhibits cellulase gene expression by interacting with the negative regulator ACE1 in Trichoderma reesei

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Communications Biology

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Abstract:
Protein methylation is a commonly posttranslational modification of transcriptional regulators to fine-tune protein function, however, whether this regulation strategy participates in the regulation of lignocellulase synthesis and secretion in Trichoderma reesei remains unexplored. Here, a putative protein methyltransferase (TrSAM) is screened from a T. reesei mutant with the ability to express heterologous -glucosidase efficiently even under glucose repression. The deletion of its encoding gene trsam causes a significant increase of cellulase activities in all tested T. reesei strains, including transformants of expressing heterologous genes using cbh1 promotor. Further investigation confirms that TrSAM interacts with the cellulase negative regulator ACE1 via its amino acid residue Arg383, which causes a decrease in the ACE1-DNA binding affinity. The enzyme activity of a T. reesei strain harboring ACE1R383Q increases by 85.8%, whereas that of the strains with trsam or ace1 deletion increases by more than 100%. By contrast, the strain with ACE1R383K shows no difference to the parent strain. Taken together, our results demonstrate that TrSAM plays an important role in regulating the expression of cellulase and heterologous proteins initiated by cbh1 promotor through interacting with ACE1R383. Elimination and mutation of TrSAM and its downstream ACE1 alleviate the carbon catabolite repression (CCR) in expressing cellulase and heterologous protein in varying degrees. This provides a new solution for the exquisite modification of T. reesei chassis.

Summary:
Introduction: The filamentous fungus Trichoderma reesei(teleomorph Hypocrea jecorina) is one of the most well-known industrial strains used for producing cellulase and hemicellulases, which release fermentable soluble sugars from plant biomass. The high production levels of lignocellulose-degrading enzymes in T. reseis dependent on induction by cellulose- and hemicellulose-containing plant polysaccharide mixtures. Besides, T.reeis also an ideal chassis for the expression of various heterologous proteins. The highly efficient expression level of target proteins may play a key role in reducing the cost of bioprocessing. Deleting and/or overexpressing transcription factors (TFs) is a straightforward method to increase the target protein yield of T. Reesi.

Results: In our previous research, a series of transformants were successfully constructed by the random insertion of the encoding gene of a -glucosidase fromAspergillus terreus, most of them (represented by the mutant atbg-A1) demonstrated significantly increased -gluscosidasese production (hydrolysis activity ofp-nitrophenyl--D-glucopyranoside,pNPGase activity) and slightly increased filter paper activity (FPA), compared to that of the parent strain Rut-C30. However, one mutant, named as atbcg-D3, unexpectedly exhibited lower filter paper activities (60 of that of Rut - C30) but much higherpNPGAseactivity (100-fold increase) (Supplementary Fig). Unexpectedly, atbbgu10 exhibited the highest glycosidic activity in inducing medium, with a 200-fold increased inpNGAseactivity compared with Rut C30 (Fig Fig. 1Evaluation of enzymatic activity and protein production by transformants expressing BglS in induction and repressing conditions.aEvaluations of glucosesidicase (pNGPasa)activities under inducining condition.cEvaluates of extracellular protein concentrations under inducting condition .dSDS-PAGE analysis of protein production under induction condition.

Discussion: ACE1 was identified as a repressor of cellulase and xylanasas expression inT. reesei. It has been testified that ACE1, XYR1, ACE2, ACE3, and others may bind to the promoters ofxyr1,cbh1and other cellulosae or hemicellulases genes competitively. C2H2-type TF ACE1.

Reporting summary: Further information on research design is available in thelinked to this article. Supplementary information Supplementary Information: . . ( ) .. (( ). .( ). (.) . ( ) - .. ( ) (- -) ( ) : , -, (), , (,) : ( ), ( )-- (,)-

Figures:
Figure 1: Evaluation of enzymatic activity and protein production by transformants expressing BglS in inducing and repressing conditions.

Figure 2: Schematic diagram of large missing fragments in atbg-D3 and atbg-U10.

Figure 3: Interaction analysis of protein methyltransferase Tr SAM and ACE1.

Figure 4: Evaluation of cellulase production of Rut-C30 and its mutants.

Figure 5: Comparison of ACE1 and ACE1 R383Q abilities to bind DNA.

Figure 6: Schematic model of Tr SAM-mediated regulatory mechanism during carbon catabolite repression in T. reesei.

Mentioned biomedical entities:

Genes: ACE1, ACE2, ACE3, BglR, BglS, CA, CBH1, Cre1, FPA, G9a, GAL4, GATA4, GST, Hda1, Lae1, MO, Mig1, MyoD, PRC2, RK, Roc1, Vel1, XYR1, ace1, bglR, cbh1, cellulase, cre1, histone, p53, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, protoplast

Organisms: E. coli, mouse

Tissues: N/A

Pathways: N/A

Query Result 22/25

Enhancement of cellulase production in Trichoderma reesei RUT-C30 by comparative genomic screening

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Microbial Cell Factories

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Keywords Frequency (TF): cellulase: 0.018; reesei: 0.012; cbh1: 0.000; xyr1: 0.001

Abstract:

Cellulolytic enzymes produced by the filamentous fungus Trichoderma reesei are commonly used in biomass conversion. The high cost of cellulase is still a significant challenge to commercial biofuel production. Improving cellulase production in T. reesei for application in the cellulosic biorefinery setting is an urgent priority.

Summary:

Background: Lignocellulosic biomass, which consists of cellulose, hemicellulose, and lignin, is a renewable resource that is abundantly available for the production of biofuels and chemicals. Biological conversion of lignocellulo sic biomass into fermentable sugars by cellulolytic enzymes is an environment-friendly and promising approach ,. However, the production cost of biomass-degrading enzymes remains a significant challenge for commercial biofuel production .

Results: SS-II, a T. reeseihyper-cellulolytic mutant strain, exhibited significantly enhanced CMCase activity, pNPCasa activity, and extracellular protein production compared with those of NG14. The same FPasease loading (20 filter paper cellula) was observed in SSII. SSiI is a strain with a high cellulolysis rate.

Conclusions: T. reesei hyper-cellulolytic strain SS-II derived from NG14 grew faster and more effectively degraded lignocellulosic biomass to produce more glucose than the RUT-C30 strain, which was also isolated from ng14. SSII genome sequencing revealed a full-length cre1 . Using comparative genomic analysis, we identified multiple mutations in the SS II strain. Among these mutated SS II genes, the tre108642 and tre56839 deletions in T.

Discussion: In this study, the genetic background of an industrial hyper-cellulase-producing strain of Trichoderma reesei SS-II, was analyzed. T. reseis SSII is derived from T. Reesis NG-14, the parental strain of the industrial precursor strain RUT-C30. The SS II strain exhibited higher CMCasa activity and increased growth compared with those of the RUT C30 strain. Cellulases produced by SS II was more efficient than that from RUT - C30 for the degradation of lignocellulosic biomass to produce glucose. However, the SS - II exhibited similar levels of FPass activity, and lower pNPCas activity compared to those of RUT. C30. Thus, we sought to combine the mutations of R. C 30 and SS III to acquire more efficient cellulased strains.

Figures:

Figure 1: Phenotypic characteristics of SS-II. a Cell line of T. reesei hyper-cellulolytic mutant SS-II. UV, ultraviolet; NTG, N-nitrosoguanidine. Biomass dry weight of T. reesei strains were measured in MA medium containing 2% (w/v) glucose (b), 2% (w/v) lactose (c), or 2% (w/v) Avicel (D) as the sole carbon source. FPase (e), CMCase (f), pNPCase (g), and pNPGase (h) activities and total secreted protein (i) of T. reesei strains were measured using Avicel as the carbon source. j Hydrolysis of pretreated corn stover using the crude enzyme from T. reesei strains at 20 FPU/g dry biomass. Values are the meanSD of the results from three independent experiments. Asterisks indicate significant differences (p0.05, p0.01, p0.001, Students t test)

Figure 2: Cellulase production of SS-II- cre1 96 . a Schematic diagram of the construction of the mutant SS-II- cre1 96 . The truncated form of the cre1 96 sequence from RUT-C30 was introduced in SS-II to replace the full-length cre1 by double crossover exchange. The hygromycin cassette was excised to form SS-II- cre1 96 . T. reesei strains were cultivated in MA medium using lactose as the carbon source. FPase activity was calculated in two types of units, U/mL (b) and IU/mg biomass (c). pNPCase activity was also calculated in two types of units, U/mL (d) and IU/mg biomass (e). Values are the meanSD of the results from three independent experiments. Asterisks indicate significant differences (p0.05, p0.01, p0.001, Students t -test)

Figure 3: Cellulase activity and secreted protein concentration of T. reesei mutants. FPAase (a), CMCase (b), pNPCase (c), pNPGase (d) activities, and total secreted protein (e) of T. reesei strains were measured using Avicel as the carbon source. Values are the meanSD of the results from three independent experiments. Asterisks indicate significant differences (p0.05, p0.01, p0.001, Students t -test)

Figure 4: Expression levels of cellulase related genes in T. reesei mutants. a Gene expression levels were evaluated by qPCR after growth in Avicel for 48, 72, and 96h. Relative expression levels of cbh1 , cbh2 , egl1 , egl2 , bgl1 , and xyr1 . b Relative expression level of hac1 gene. Data were normalized to the expression of the RUT-C30 at 48h for each tested gene, and sar1 expression was used as an endogenous control in all samples. Values are the meanSD of the results from three independent experiments. Asterisks indicate significant differences (p0.05, p0.01, p0.001, Students t -test)

Mentioned biomedical entities:

Genes: ACE1, ACE2, Arg, BglR, CA, CRE1, CRZ1, Cellulase, Co, Crt1, GRD1, LAE1, MA, MFS, MO, PCS, Pac1, QD, Rce1, Stp1, XYR1, bgl1, cbh1, cbh2, cellulase, cre1, egl1, grd1, hac1, sar1, vib1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: Cell, Cell line

Organisms: Escherichia coli

Tissues: N/A

Pathways: N/A

Query Result 23/25

Synergistic and Dose-Controlled Regulation of Cellulase Gene Expression in Penicillium oxalicum, bRegulation of Cellulase Gene Expression

Li, Zhonghai; Yao, Guangshan; Wu, Ruimei; Gao, Liwei; Kan, Qinbiao; Liu, Meng; Yang, Piao; Liu, Guodong; Qin, Yuqi; Song, Xin; Zhong, Yaohua; Fang, Xu; Qu, Yinbo; Glass, N. Louise

PLoS Genetics

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Abstract:

Filamentous fungus Penicillium oxalicum produces diverse lignocellulolytic enzymes, which are regulated by the combinations of many transcription factors. Here, a single-gene disruptant library for 470 transcription factors was constructed and systematically screened for cellulase production. Twenty transcription factors (including ClrB, CreA, XlnR, Ace1, AmyR, and 15 unknown proteins) were identified to play putative roles in the activation or repression of cellulase synthesis. Most of these regulators have not been characterized in any fungi before. We identified the ClrB, CreA, XlnR, and AmyR transcription factors as critical dose-dependent regulators of cellulase expression, the core regulons of which were identified by analyzing several transcriptomes and/or secretomes. Synergistic and additive modes of combinatorial control of each cellulase gene by these regulatory factors were achieved, and cellulase expression was fine-tuned in a proper and controlled manner. With one of these targets, the expression of the major intracellular -glucosidase Bgl2 was found to be dependent on ClrB. The Bgl2-deficient background resulted in a substantial gene activation by ClrB and proved to be closely correlated with the relief of repression mediated by CreA and AmyR during cellulase induction. Our results also signify that probing the synergistic and dose-controlled regulation mechanisms of cellulolytic regulators and using it for reconstruction of expression regulation network (RERN) may be a promising strategy for cellulolytic fungi to develop enzyme hyper-producers. Based on our data, ClrB was identified as focal point for the synergistic activation regulation of cellulase expression by integrating cellulolytic regulators and their target genes, which refined our understanding of transcriptional-regulatory network as a seesaw model in which the coordinated regulation of cellulolytic genes is established by counteracting activators and repressors.

Summary:

Introduction: Cellulolytic fungi have an inherent characteristic of cellulose deconstruction and can be used for bioconversion of insoluble plant cell wall polysaccharides into fermentable sugars. However, incomplete knowledge of transcriptional regulatory networks for cellulolysis has hampered the systematic improvement of cellulase production. These cellulolitic system genes are coordinately but differentially regulated in various cellullase producers. Further characterization and manipulation of the cellululases regulatory networks components will allow the rational engineering of cellulose fungus for improved enzyme production.

Results: We first sought to identify the transcription factors that play roles in cellulolytic gene expression systematically. We first amplification of the flanking sequences of these TF-disrupting cassettes. We selected three colonies per gene from these resulting transformants. The conidia from these primary transformants were purified by repeating the mono-spore isolation twice on the pyrithiamine resistance plates to obtain homokaryotic knockout mutants. The transcription factor gene replacement withptrawas verified by PCR-based screening.

Discussion: Several conserved and essential transcription factors for cellulase and hemicellulases genes have been recently described in cellulolytic fungi; yet, the regulatory patterns of these factors have not been thoroughly analyzed . In this study, a single-gene disruptant mutant library of the transcription regulators in P. oxalicumwas constructed, and 20 transcription factors that play a pivotal role in the activation or inhibition of cellulose deconstruction were identified. Of these screened transcription factors, ClrB, CreA, XlnR, and AmyR were selected for further analyses. The cellullase expression regulated by additive cellulotic effectors was observed. The results correspondingly provided comprehensive information for deciphering and redesigning a cellululased expression regulation network for rational engineering of cellululase hyper-producers.

Conclusion: Cellulase formation apparently occurred because of consistent respective regulators, including the characterized or novel transcription factors identified in P. oxalicum in this context. Using this model, the accumulation of intracellular cellodextrins can trigger signaling cascades that include expression of cellulases genes repressed by CreA and AmyR and activated by ClrB and XlnR. However, our data also support that the transcriptional regulation for CreA, AmyR, Clr B and xln R genes is a powerful part of the regulatory network of cellulase gene expression. In the early cellulolytic induction, ClreB functions to repress expression of AmyR , whose expression level is activatedby CreA. Moreover, transcriptional expression for Clrb, Amy R, XrnR and Clr-B-2 genes is also significantly repression by CCR mediated by Crea in the presence of glucose. The data established ClrA as a focal point to regulate cellullasae expression by integrating other regulators and the target genes of these regulators. These observations also suggested the hypotheses that the rational design of cellular or high-value protein super producers might be guided in the future for the combinatorial effects of diverse cellulolitic effectors.

Figures:

Figure 1: Phenotypes on different carbon sources and enzyme activities of culture supernatants from mutants containing clrB , xlnR and/or creA mutations.

Figure 2: Regulation of cellulolytic gene expression by ClrB, XlnR and CreA.

Figure 3: Defining the categories of genes potentially regulated by ClrB.

Figure 4: Enzyme activities of culture supernatants from mutants containing overexpression of clrB and/or xlnR on cellulose.

Figure 5: CreA regulated clrB , clrB-2 , xlnR , and amyR expression and caused defects in microscopic phenotypes.

Figure 6: Heatmap of cellulose regulons and protein profiles of culture supernatants from P . oxalicum strains containing mutations of clrB or/and creA genes.

Figure 7: Lack of Bgl2 and overexpression of ClrB or lack of CreA additively induced cellulase expression.

Figure 8: Overexpression of clrB or xlnR in triple-mutant RE-10 enhanced hydrolase expression under inducer-inducing conditions.

Figure 9: Lack of AmyR decreased amylase activity and increased cellulase expression on cellulose.

Figure 10: Expression of amyR was regulated by ClrB and CreA, and hierarchical clustering of RPKM for amyR regulon genes on cellulose.

Figure 11: Electrophoretic mobility shift assay and protein interaction among ClrB, CreA, AmyR and XlnR.

Figure 12: Model of regulation of cellulase expression by ClrB, CreA, AmyR, XlnR and ClrB-2 in P . oxalicum .

Mentioned biomedical entities:

Genes: 67, Ace1, AmyR, B, B8, Bgl1, Bgl2, BglR, CLR-1, CRE1, CdtC, Cellulase, ClrB, Cre-1, CreA, FPA, GST, ManR, Tm, Trp, XYR1, XlnR, actin, amyR, amylase, beta-1, bgl1, bgl2, cbh, cbh1, ccg-1, cellulase, clrB, cre1, creA, gluA, gpdA, xlnR, xyn1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, protoplast

Organisms: A. nidulans, Escherichia coli

Tissues: Ion, lens, tissue, tube

Pathways: N/A

Query Result 24/25

Novel genetic tools that enable highly pure protein production in Trichoderma reesei

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Scientific Reports

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Keywords Frequency (TF): cellulase: 0.002; reesei: 0.006; cbh1: 0.003; xyr1: 0.000

Abstract:

Trichoderma reesei is an established protein production host with high natural capacity to secrete enzymes. The lack of efficient genome engineering approaches and absence of robust constitutive gene expression systems limits exploitation of this organism in some protein production applications. Here we report engineering of T. reesei for high-level production of highly enriched lipase B of Candida antarctica (calB) using glucose as a carbon source. Multiplexed CRISPR/Cas9 in combination with the use of our recently established synthetic expression system (SES) enabled accelerated construction of strains, which produced high amounts of highly pure calB. Using SES, calB production levels in cellulase-inducing medium were comparable to the levels obtained by using the commonly employed inducible cbh1 promoter, where a wide spectrum of native enzymes were co-produced. Due to highly constitutive expression provided by the SES, it was possible to carry out the production in cellulase-repressing glucose medium leading to around 4 grams per liter of fully functional calB and simultaneous elimination of unwanted background enzymes.

Summary:

Introduction: T.reesei has also been proposed to provide a low-cost production platform for biopharmaceuticals, such as interferons. Some studies have reported total secreted protein titers exceeding 100g/L, which highlight the productivity potential of this organism for protein production applications. The filamentous fungus Trichoderma reeesi T re esai is an important and widely exploited organism for production of industrial enzymes with a high, natural capacity to secrete cellulolytic enzymes. This property has made it an attractive host for multiple industrial applications,, particularly for production .

Results: Development of an efficient CRISPR/Cas9 transformation method for Cas9 nucleo-protein complexes to enable multiplex gene deletions/replacements inT. reesei. We tested simultaneous deletion of three major cellulase genes: cellobiohydrolase II (cbh2), endoglucanasasi I (egl1), and endogl2 (el2).

Discussion: CRISPR/Cas9 has been successfully established in T.reesei, where up to three simultaneous genes deletions were demonstrated in a strain carrying a Cas9 expression cassette integrated in the genome. To eliminate the need for prior strain engineering, we employed the Cas 9 protein andin vitrogRNA transformation protocoland showed feasibility of the approach for simultaneous triple deletions with a 12 frequency (Table). This, to our knowledge, is the highest frequency for triple deletion reported with CRISPERCaS9 in T reeesi.

Figures:

Figure 1: Scheme of the CRISPR/Cas9-mediated multiplexed gene deletions in T . reesei . The donor DNA molecules were constructed, each containing the pyr4 selection marker gene flanked by 1000bp of DNA region homologous to 5-upstream and 3-downstream regions of the selected gene. Two gRNAs were designed to target Cas9 protein for introducing double stranded break at the 5- and 3-ends of each targeted coding sequence. A single-transformation was performed combining 6 in vitro pre-assembled Cas9 ribonucleo-complexes, three donor DNAs, and the T . reesei protoplasts. Simultaneous deletion of the three cellulase genes, cellobiohydrolase II (cbh2), endoglucanase I (egl1), and endoglucanase II (egl2), was achieved in about 12% of the resulting colonies (Table 1).

Figure 2: Comparison of approaches for the calB production in T . reesei . (A) Scheme of the synthetic expression system used for the CBHI-calB production. The expression of synthetic transcription factor (sTF; Bm3R1-VP16) was driven by the hfb2 core promoter (T . reesei origin), which provides a low and constitutive sTF expression via its basal transcription activity. The sTF recognizes the binding sequences (8BS) in an engineered promoter of the cbh 1- calB gene. The sTF binding sites are followed by the An201 core promoter (A . niger origin), forming the synthetic promoter. The sTF expression cassette was integrated in a single copy into the cbh2 locus and the CBHI-calB expression cassette was integrated as three copies in the cbh1 , egl1 and egl2 loci. (B) Scheme of a control CBHI-calB expression cassette. The cbh1 - calB gene was expressed from the commonly used cellulose inducible cbh1 promoter. The cassette was integrated into cbh1 locus in a single copy. (C) The SDS-PAGE analysis of proteins produced (secreted) into culture media of bioreactor cultivations. The strains carrying expression cassettes shown in (A) (SES strain) and (B) (cbh1 p strain) were cultivated either in SG-lactose medium or in glucose-containing medium. The cultivation media samples were collected as indicated and diluted 1:10 in water prior to analysis. Purified, full length CBHI protein was used as a loading standard and the proteins were visualized by Coomassie staining. (D) The total protein concentrations (blue bars, left y-axis) and the calB lipase activities (orange bars, right y-axis) in the samples collected on day 5 of the bioreactor cultivations. The lipase activity levels were normalized to total protein concentrations and the results are presented as a percent of commercially available calB activity. The values and the error bars represent means and standard deviations from three

technical replicates.

Figure 3: Production of highly enriched calB protein using a short secretion signal sequence. (A) Scheme of the SES system used for the production of calB with a short secretion signal sequence (SS; 26 N-terminal amino-acids from A . awamori -amylase). The sTF expression cassette was integrated into the egl1 locus (single copy) and the AaSS-calB expression cassette was integrated into the cbh1 , cbh2 and egl2 loci (triple copy). (B) The SDS-PAGE analysis of proteins produced (secreted) into culture media of bioreactor cultivations. The strain producing CBHI-calB (SES strain in Fig. 2) and the AaSS-calB strain carrying the expression cassettes from (A) were cultivated either in SG-lactose medium for 7 days or in glucose-containing medium for 5 days. The cultivation media samples were collected as indicated and diluted 1:30 in water prior to analysis. Commercially available calB protein was used as a loading standard and the proteins were visualized by Coomassie staining. The samples from the last day of each cultivation subjected to gel-filtration chromatography and freeze-drying were also analyzed after reconstitution in water and 1:30 dilution (marked with asterisk). (C) The total protein concentrations (blue bars, left y-axis) and the calB lipase activities (orange bars, right y-axis) in the samples collected from the bioreactor cultivations and in the gel-filtrated, freeze-dried samples (marked with asterisk). The lipase activity levels were normalized to total protein concentrations and the results are presented as a percent of commercially available calB activity. The values and the error bars represent means and standard deviations from three technical replicates. (D) Transcription analysis of strains cultivated in bioreactors. The calB and sTF transcript levels were normalized to the transcript levels of native ubc gene. Transcript levels of the sTF are shown on the secondary (right) y-axis. The values and the error bars represent means and standard deviations from two biological (four technical) replicates.

Mentioned biomedical entities:
Genes: 9.5, Cas9, ER, Kex2, XYR1, act1, calB, cbh1, cbh2, cellulase, egl1, nsf1, pdi1, pyr4, sTF, ubc, xyn1, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: A. awamori, E. coli

Tissues: tube

Pathways: N/A

Query Result 25/25

Redesigning transcription factor Cre1 for alleviating carbon catabolite repression in Trichoderma reesei

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Abstract:

Carbon catabolite repression (CCR), which is mainly mediated by Cre1 and triggered by glucose, leads to a decrease in cellulase production in Trichoderma reesei. Many studies have focused on modifying Cre1 for alleviating CCR. Based on the homologous alignment of CreA from wild-type Penicillium oxalicum 1142 (Po-0) and cellulase hyperproducer JUA10-1(Po-1), we constructed a C-terminus substitution strainPo-2with decreased transcriptional levels of cellulase and enhanced CCR. Results revealed that the C-terminal domain of CreAPo1 plays an important role in alleviating CCR. Furthermore, we replaced the C-terminus of Cre1 with that of CreAPo1 in T. reesei (Tr-0) and generated Tr-1. As a control, the C-terminus of Cre1 was truncated and Tr-2 was generated. The transcriptional profiles of these transformants revealed that the C-terminal chimera greatly improves cellulase transcription in the presence of glucose and thus upregulates cellulase in the presence of glucose and weakens CCR, consistent with truncating the C-terminus of Cre1 in Tr-0. Therefore, we propose constructing a C-terminal chimera as a new strategy to improve cellulase production and alleviate CCR in the presence of glucose.

Summary:

Introduction: The abundant natural material lignocellulose, which is converted into fermentable sugars or other chemicals through enzymatic hydrolysis, is considerably under-utilized ,. The biotransformation of lignocellulocellulose is significantly hindered by the high cost and the low yield of cellulase . In T. reesei, carbon catabolite repression (CCR), which is present in nearly all heterotrophic hosts ,, is triggered to downregulate cellulolytic enzyme production in the presence of glucose. The protein Cre1 (encoded by cre1 in T.

Results: CreA fromP. oxalicum. The amino acid sequences of CreA between Po-0 and Po-1 were compared using ClustalX2 and then the sequences at the C-terminus of CreAPo1greatly differed from those of the original CreAP0(a). It was obvious that a frameshift mutation at site 1205 of nucleotide sequence of Cre0resulted in the amino acid splicing of Cre1 and CreAP1within the C terminus. To investigate the effect of this frameshift mutant on the regulation of cellulase expression, the C -terminal sequence ofCreAPo2was substituted with the C sequence ofcreAPo0extending from aa 401 to ai 417, thereby generating a new chimera transcription factor named CreAP2(a) and CreA(a-1)was recombined into Po-1, generating the transformant Po-2.Cel7ais a major cellulose-encoding gene, andxlnRis the gene encoding transactivator inP.oxalici. Subsequently, the expression levels ofcel7ain Po-2 were sharply decreased using glucose as the sole carbon source, but it was unchanged using avicel as the only carbon source. It was therefore suggested that the Cterminal frameshiftmutation of Crea alleviated CCR, resulting in cellullase hyperproduction.

Discussion: TEXT: The CCR effect facilitates the preferential assimilation of energy-efficient and readily available carbon sources such as glucose or xylose by inhibiting the expression of enzymes involved in the catabolism of other carbon sources ,. Therefore, many attempts have been made to suppress CCR and/or its effect to improve the expression . Nakari-Setl et al. reported that the deletion of cre1 increased the quantity of cellulases produced by the wild-type T. reesei QM6a strain. However, it was also reported that deletion of Cre1 did not result in inceased total cellullase production in T.

Conclusions: Based on the findings of the present study, we suggest that phosphorylation of the C-terminus of CreA/Cre1 is one of the reasons driving CCR. We demonstrated that constructing a C-termal chimera of Cre1 in T. reeseiimproved the transcription levels of cellulase and alleviated CCR in the presence of glucose. Notably, we proved that phosphatylation of the c-terminal of Cre1/Cre2 plays an important role on its subcellular localization and has a direct relationship with CCR and providing a basis for Cre1 phosphorylation-based research involved in CCR

Figures:

Figure 1: Amino acid sequence alignments and in silico domain prediction of Cre1/CreA. (a) Amino acid sequence alignments of Cre1/CreA from Penicillium oxalicum Po-0, P. oxalicum Po-1, and Trichoderma reesei Tr-0. The C-terminus of Cre1/CreA is marked with a red box. (b) In silico domain prediction of Cre1. Numbers indicate the amino acid (aa) positions, and colored boxes indicate identified C 2 H 2 zinc finger domains (gray); pink, nuclear localization signal (NLS); light blue, transactivation domain (TAD); yellow, Q (glutamine); dark blue aspartic (D) and glutamic acid (E); green, Q-X7- Q-X7-Q; violet, nuclear export signal (NES); orange, repression domain; and blank, the C-terminus of Cre1/CreA. The putative domains of Cre1 were predicted by a number of in silico prediction tools and alignment algorithms as described in the Materials and Methods, section 2.3 . (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Figure 2: Comparison of the transcriptional levels among transformant Po-2 and parent strains Po-1 in P . oxalicum . Relative gene expression of cbh1 and xlnR between Po-1 (blank) and its complemented strain Po-2 (dark gray) grown on culture using (a) glucose or (b) avicel as the sole carbon

source. Gene expression levels were normalized (logarithm-2 CT analysis) to that of the reference sample Po-1 using the reference gene -actin . Mean values are presented; error bars indicate standard deviation from three independently grown cultures.

Figure 3: Transcript levels of cel7a and xyr1 in the transformant strains Tr-1 and Tr-2. The transcriptional levels of cel7a and xyr1 between Tr-0 (blank) and its mutated strains Tr-1 (gray), Tr-2 (dark gray) grown in culture using (a) glucose or (b) avicel as the sole carbon source. Gene expression levels were normalized (2 CT analysis) to that of -actin gene. Mean values are presented; error bars indicate standard deviation from three independently grown cultures.

Figure 4: Analysis of phenotypic characterization, transcriptional levels and cellulase activities. (a) Morphologies of the parent strain and transformants on plates containing different carbon sources. For growth assays, transformants and parent strain were grown on plates with 2% (w/v) potato dextrose agar (PDA), avicel, glucose, or glycerol as a sole carbon source. Plates were incubated at 30C and pictures were taken after 144h. (b) Microscopic observation of hyphae of parent strain and transformants. Approximately, 110 3 spores were inoculated on slides with solidified medium containing PDA, or MM with 2% glucose at 30C for 48h. (c) Transcriptional levels of cel7a and xyr1 in Tr Cre1 5M (gray) and parent strain (blank) grown in presence of glucose (left) and avicel (right) as carbon source. Gene expression levels were normalized (2 CT analysis) to that of actin. Mean values are shown; error bars indicate the standard deviation of three independently grown cultures. (d) T. reesei transformants and the parent strain were cultivated in liquid medium supplemented with 2% (w/v) glucose (left) or avicel (right) as carbon source. Activity of p -nitrophenyl--D-cellobioside (p NPCase), filter paperase (FPase) activity and soluble protein were measured in biological and technical duplicates. Enzymatic activities are given as mean values, with error bars indicating the standard deviation. Symbols: blank, parent strain; dark gray, Tr Cre1 5M .

Figure 5: Fluorescence microscopy analysis of Cre1-GFP and Cre1 5M -GFP. The nuclei were stained with Hoechst 33258. Scale bar=20m.

Mentioned biomedical entities:
Genes: Cel7a, Clr1, Clr2, Cre1, CreA, E100, SRR, Sen, Tr-1, Xyr1, actin, cbh1, cellulase, cre1, xlnR, xyr1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: T. reesei

Tissues: N/A

Pathways: N/A